

Script: Analysis I

“Young man, in mathematics you don’t understand things. You just get used to them”
John Von Neumann

Introduction

Hello dear reader, and welcome to this amazing journey into one of the most fundamental subjects of modern mathematics, the one of Analysis. If you are here, reading these few lines, you may probably be having a tough time, struggling with the preparation for the exam session, and, very likely, facing some new difficulties for the first time. No panic! These notes, or, if you prefer, this script, are meant to help you navigate this hard time of yours, hopefully alleviating the stress that, we know, the study of a subject like Analysis I can carry with itself. Moreover, we really hope that, despite the difficulties, you may also realize how beautiful it can be to explore the world of mathematics, and, more specifically, the very special realm of that world that this subject belongs to. It is indeed in this spirit that we wrote the following, in the hope you may share at least part of the enthusiasm and astonishment that we have genuinely experienced during the writing.

This piece of work has been realized in the context of a VPK course on Analysis I, organized by VMP association, at ETH Zürich, in the Fall Semester of 2023. In particular, it is meant to (in a reasonably consistent way) reproduce, both in the content and the order of its presentation, the notes of the course *Analysis I: One Variable*, taught by Professor Alessio Figalli. Yet, every error in both the formulation of the results and their presentation, as well as all the typos are to be ascribed to the authors of the script only. Moreover, we deem it important to underline that these notes are to be thought neither exhaustive nor complete, but instead, they are supposed to work as a guide, or, if you like, a friend, to rely on during the preparation for the exam. Here, you will not find proof of the theorems discussed in the lectures, but only the statements. You will not find exercises, but some questions aimed at stimulating your critical thinking and pushing forward the comprehension of the subject. So, turn the page and have fun, my friend!

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Chapter 1

Sets and a Basic Toolbox

This first section must be intended as a staple ingredient of the course of Analysis I. Since one of the most popular formulations of modern mathematics relies on set theory and hence on the notion of set, we now introduce this fundamental object and dwell on some of its properties and operations that it allows to construct. If you are interested in foundational questions, try to type “Zermelo-Fraenkel set theory” into some search engine online, and have fun!

First, we need to notice that a set is not a formally defined object in mathematics. It is rather a primitive concept, and similarly to what we do with axioms, we must be content with what our intuition provides. A set is a mathematical model for a collection of objects. These objects, once seen as being in the set, are called **elements**. Sets are usually denoted by capital letters, like $A, B, \dots$, or S . If a is an element of the set A , we will write $a \in A$. As we shall soon see, sets can be classified in many different ways, depending on the number of their elements, the “nature” of their elements, how their elements interact with each other, and so on. When we speak of the number of elements in a set, we refer to it as its **cardinality**.

1.1 Operations with Sets

It is possible to define operations between sets, as we can see from the following definition.

Definition 1.1.1: Set Operations

Let P and Q be two sets. The **intersection** $P \cap Q$, the **union** $P \cup Q$, the **relative complement** $P \setminus Q$, and the **symmetric difference** $P \Delta Q$ are defined by

$$P \cap Q := \{x | x \in P \text{ and } x \in Q\}, \quad (1.1)$$

$$P \cup Q := \{x | x \in P \text{ or } x \in Q\}, \quad (1.2)$$

$$P \setminus Q := \{x | x \in P \text{ and } x \notin Q\}, \quad (1.3)$$

$$P \Delta Q := (P \cup Q) \setminus (P \cap Q). \quad (1.4)$$

The operations of union and intersection are easily extended to any family of sets.

Definition 1.1.2: Union and Intersection of several sets

Let $\mathcal{A}$ be a family of sets. Then we define the **union** and the **intersection** of the sets in $\mathcal{A}$ as

$$\bigcup_{A \in \mathcal{A}} A := \{x | \exists A \in \mathcal{A} : x \in A\}, \quad \bigcap_{A \in \mathcal{A}} A := \{x | \forall A \in \mathcal{A} : x \in A\}. \quad (1.5)$$

Note that we did not make any assumptions on the cardinality of $\mathcal{A}$, which means that the intersection and the union are defined even for infinite families of sets.

1.1.1 De Morgan's Laws

Given two sets A and B , we say that A is a **subset** of B , and we write $A \subseteq B$, if the following implication holds:

$$x \in A \implies x \in B.$$

Given a set X , and a subset $A \subseteq X$, we define the **complement** of A in X as

$$A^c := \{x \in X \mid x \notin A\}.$$

It is easy to see that $A^c = X \setminus A$.

Theorem 1.1.1: De Morgan's Laws

Let X be a set, and let $A, B \subseteq X$ be two subsets. Then, the following equalities hold:

1. $(A \cup B)^c = A^c \cap B^c$;
2. $(A \cap B)^c = A^c \cup B^c$.

More generally, if $(A_i)_{i \in I}$ is some family of subsets of X , for some index set I , we have

$$\left(\bigcup_{i \in I} A_i \right)^c = \bigcap_{i \in I} A_i^c, \quad \left(\bigcap_{i \in I} A_i \right)^c = \bigcup_{i \in I} A_i^c.$$

Chapter 2

The Real Numbers: Maximum, Supremum, and Sequences

2.1 The Axioms of the Real Numbers

As you may have already realized, the language of mathematics is quite peculiar, and full of symbols that are not familiar to one that does not know this world from the inside. Along your journey into this beautiful realm, you will get used to this new language, and sometimes notice that there is no universal agreement on the notation. In order to improve the readability of this script, we need to introduce explicitly a convention we intend to follow in the rest of the notes. We will write $\mathbb{N}$ to denote the set of all non-negative integers, namely

$$\mathbb{N} := \{0, 1, 2, 3, \dots\}.$$

It is worth making this clear since some authors use the same symbol to denote the set of all the positive integers, thus not containing the number 0. Moreover, we will use the convention

$$\mathbb{R}^+ := \{x \in \mathbb{R} \text{ such that } x > 0\}, \quad \mathbb{R}_0^+ := \mathbb{R}^+ \cup \{0\}.$$

2.1.1 Some basics structures

In this chapter, we aim to provide a first overview of one of the main objects under our consideration, namely the field of real numbers. If on the one hand, we may already have an intuitive notion of real numbers, it's now the right time to introduce some rigor and formalism.

In the first chapter, we said that a set can be classified in many different ways. An entire branch of mathematics, namely what we call algebra, deals with classification criteria that take into account the properties of the operations defined for a given set. Let's explore some of the basic structures that these criteria identify.

Definition 2.1.1: Groups

A **group** is a (non-empty) set G endowed with an operation “ $\star$ ” that satisfies the following properties:

- Associativity;
- Neutral element;
- Inverse element.

Definition 2.1.2: Rings and Fields

A **ring** is a (non-empty) set R in which we can “add” and “multiply” elements in a compatible way. More precisely, a ring is a commutative group R whose first operation “+” is called “addition”, and it is also equipped with an additional operation called “multiplication” and denoted by “ $\cdot$ ”.

The neutral element for the addition is denoted by 0. The multiplication is associative, has a neutral element (denoted by 1), and satisfies the Distributivity property.

If $\cdot$ is also commutative, we call the ring a **commutative ring**.

A commutative ring in which every non-zero element has an inverse for the multiplication is called a **field**.

Before giving the following definition, let’s consider two sets, namely X and Y . We denote the **Cartesian product** of these two sets by $X \times Y$, which is the set defined by

$$X \times Y := \{(x, y) | x \in X, \text{ and } y \in Y\}.$$

Definition 2.1.3: Relations

Let X be a set. A **relation** on X is a subset $\mathcal{R} \subseteq X \times X$.

If $\sim$ is a relation (i.e., $x \sim y \Leftrightarrow (x, y) \in \mathcal{R}$), we call $\sim$

1. **Reflexive** if for all $x \in X$: $x \sim x$.
2. **Transitive** if, for all $x, y, z \in X$: $x \sim y, y \sim z \implies x \sim z$.
3. **Symmetric** if, for all $x, y \in X$: $x \sim y \implies y \sim x$.
4. **Antisymmetric** if, for all $x, y \in X$: $x \sim y$ and $y \sim x \implies x = y$.

A relation is called an **equivalence relation** if it satisfies 1., 2., and 3. A relation is called an **order relation** if it satisfies 1., 2., and 4.

Let’s now put together what we’ve seen so far to define a new structure, which makes us closer to the one of the real numbers.

Definition 2.1.4: Ordered Field

Let K be a field, and let $\leq$ be an order relation on the set K . We call the pair $(K, \leq)$ (or, shortly K) an **ordered field** if the following conditions are satisfied:

1. Linearity of the order.
2. Compatibility with the addition.
3. Compatibility with the multiplication.

2.1.2 Axiom of Completeness

The basic idea that motivates what follows is that “ordered fields are generally unsuitable because you can have gaps”. Since we are interested in making sense of concepts like continuity, limit, and many others, we need to fix the problem of “gaps”.

Definition 2.1.5: Completeness Axiom

Let $(K, \leq)$ be an ordered field. We say that K is **complete** or a **completely ordered field** if the following statement is true.

- Let X, Y be non-empty subsets of K such that for all $x \in X$ and $y \in Y$ the inequality $x \leq y$ holds. Then there exists $c \in K$ lying between X and Y , in the sense that for all $x \in X$ and $y \in Y$, we have $x \leq c \leq y$.

Definition 2.1.6: Real Numbers

We call **field of real numbers** any completely ordered field, which we then denote by $\mathbb{R}$.

We'd better spend a couple of words on this definition, given the importance of the real numbers in our discussion.

As often in math, defining an object is not enough, in the sense that the definition is not in itself a guarantee that the defined object exists. Indeed, it is still not clear if a field like the one we have just defined exists. We don't dwell on the details here, but what we need to know is that such a field exists. We can say even more: up to certain identifications, this field is unique. From now on, we then assume that a field of real numbers exists and is unique, thus we refer to it as *the* field of real numbers.

2.1.3 Some basic notions of Topology

In this section, we introduce some basic notions of a branch of math which is called topology. Since our focus is the field of real numbers, we here sacrifice the generality in exchange for some clarity: we thus don't say what a topology or a topological space is. Nevertheless, we may use some expressions that would eventually be more familiar to the reader who came back here later in their studies.

Definition 2.1.7: Intervals

Let $a, b \in \mathbb{R}$. We define:

- the **closed interval** $[a, b]$ as

$$[a, b] = \{x \in \mathbb{R} | a \leq x \leq b\}.$$

- The **open interval** (a, b) as

$$(a, b) = \{x \in \mathbb{R} | a < x < b\}.$$

- The **half-open intervals** $[a, b)$ and $(a, b]$ as

$$[a, b) = \{x \in \mathbb{R} | a \leq x < b\}, \text{ and } (a, b] = \{x \in \mathbb{R} | a < x \leq b\}.$$

- The **unbounded closed intervals** $[a, \infty)$ and $(-\infty, b]$ as

$$[a, \infty) = \{x \in \mathbb{R} | a \leq x\}, \text{ and } (-\infty, b] = \{x \in \mathbb{R} | x \leq b\}.$$

- The **unbounded open intervals** (a, ∞) and $(-\infty, b)$ as

$$(a, \infty) = \{x \in \mathbb{R} | a < x\}, \text{ and } (-\infty, b) = \{x \in \mathbb{R} | x < b\}.$$

Note that, in general, any bounded interval can even be empty: for example, if $b < a$, then the interval (a, b) is the empty set.

Definition 2.1.8: Neighbourhoods

Let $x \in \mathbb{R}$. A **neighborhood** of x is a set containing an open interval I such that $x \in I$. Given $\delta > 0$, the open interval $(x - \delta, x + \delta)$ is called the **δ -neighbourhood** of x .

Given a real number $x \in \mathbb{R}$, we define the **absolute value** of x as

$$|x| := \max(x, -x). \quad (2.1)$$

We'll see that, given two numbers $x, y \in \mathbb{R}$, the absolute value $|x - y|$ can be interpreted as the distance between the two numbers under consideration. With this notation, the δ -neighborhood of x can be rewritten as

$$(x - \delta, x + \delta) = \{y \in \mathbb{R} \text{ such that } |x - y| < \delta\}.$$

In words, this set collects all (and only) the real numbers that are no farther than δ from x .

Let's now introduce a core concept of Topology, the one of "open set".

Definition 2.1.9: Open and Closed Sets

A subset $U \subseteq \mathbb{R}$ is called **open** in $\mathbb{R}$ if for every $x \in U$ there exists an open interval I such that $x \in I$ and $I \subseteq U$.

A subset $F \subseteq \mathbb{R}$ is called **closed** in $\mathbb{R}$ if its complement $\mathbb{R} \setminus F$ is open.

Intuitively, a subset is open if, for any point x in the set, all points close enough to x are also in the set. Contrary to conventional usage, “open” is not the opposite of “closed”. A useful exercise is the one to show that any countable union of open sets is still open and any finite intersection of open sets is open. What can one say about the closed sets?

2.2 Complex numbers

2.2.1 The definition

If $\mathbb{R}$ is the field of real numbers, we can define the set of complex numbers as

$$\mathbb{C} := \mathbb{R}^2 := \{(x, y) \text{ such that } x, y \in \mathbb{R}\}.$$

For $(x, y) \in \mathbb{C}$, we use the convention

$$z = x + iy,$$

where i is the **imaginary unit**, x is called the **real part** of the complex number z , and y is called the **imaginary part** of z .

By defining the operations of addition and multiplication by, respectively

- $(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$
- $(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1),$

we can equip the set $\mathbb{C}$ with a nice structure, as stated in the next proposition.

Proposition 2.2.1: $\mathbb{C}$ is a Field

The set $\mathbb{C}$, together with the zero element $(0, 0)$, the one element $(1, 0)$, and the operations introduced above, is a field.

By representing the complex numbers in the **complex plane** or **Gaussian number plane**, one can appreciate the geometric meaning of the operations introduced so far and of the ones we'll see later in this section.

Definition 2.2.1: Complex Conjugation

Let $z = x + iy \in \mathbb{C}$. We call $\bar{z} := x - iy$ the complex number **conjugate** to z . The mapping $z \mapsto \bar{z}$ is called **complex conjugation**.

Can you see what means to take the complex conjugate of a number $z \in \mathbb{C}$ from a geometric perspective? The real part is still the same, but the imaginary part has changed its sign. Hence, taking the complex conjugate amounts to reflect the point $z \in \mathbb{C}$ with respect to the horizontal axis.

Lemma 2.2.1: Properties of Complex Conjugation

The complex conjugation satisfies the following properties:

1. *For all $z \in \mathbb{C}$, we have $z\bar{z} \in \mathbb{R}$ and $z\bar{z} \geq 0$. Furthermore, for all $z \in \mathbb{C}$, the equality $z\bar{z} = 0$ holds if and only if $z = 0$.*
2. *For all $z, w \in \mathbb{C}$, we have $\overline{w + z} = \bar{w} + \bar{z}$.*
3. *For all $z, w \in \mathbb{C}$, we have $\overline{w \cdot z} = \bar{w} \cdot \bar{z}$.*

2.2.2 The Absolute Value on the Complex Numbers

We earlier mentioned the concept of absolute value, but we then were in the context of real numbers. We now define it in a precise way in a more general setup, the one of complex numbers. The following definition is thus meaningful also for real numbers.

Definition 2.2.2: Absolute Value

The **absolute value** or **norm** on $\mathbb{C}$ is the function $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$ given by

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2},$$

where $z = x + iy$.

Do you remember what we said when we first spoke of the absolute value? We then related it to the concept of distance. Try to visualize the number $|x - y|$ for some $x, y \in \mathbb{C}$ with the help of the complex plane.

Even though we here use the term “distance” in a loose fashion, it’s worth noticing that there exists a rigorous mathematical definition of it (which leads to the concept of metric space). One can show that the absolute value, according to such a definition, is indeed a distance, and, among the other properties, satisfies the so-called **triangle inequality**.

Proposition 2.2.2: Triangle Inequality

For all $w, z \in \mathbb{C}$, the inequality $|z + w| \leq |z| + |w|$ holds.

Proposition 2.2.3: Cauchy-Schwarz Inequality

For all $z = x_1 + iy_1, w = x_2 + iy_2 \in \mathbb{C}$, we have

$$x_1x_2 + y_1y_2 \leq |z||w|.$$

If you caught what we meant by suggesting interpreting the absolute value as a distance, the next definition along with the particular words used in it won’t be of surprise.

Definition 2.2.3: Circular Disks

The **open circular disk** with radius $r > 0$ around $z \in \mathbb{C}$ is the set

$$B(z, r) := \{w \in \mathbb{C} \text{ such that } |z - w| < r\}.$$

The **closed circular disk** with radius $r > 0$ around $z \in \mathbb{C}$ is the set

$$B(\bar{z}, r) := \{w \in \mathbb{C} \text{ such that } |z - w| \leq r\}.$$

As $\mathbb{R}$, the field $\mathbb{C}$ can be equipped with a topology and thus the concepts of open and closed set still make sense in the context of complex numbers.

Definition 2.2.4: Open and Closed Sets

A subset $U \subseteq \mathbb{C}$ is called **open** in $\mathbb{C}$ if for every point in U there exists an open circular disk around that point contained in U .

A subset $A \subseteq \mathbb{C}$ is called **closed** in $\mathbb{C}$ if its complement $\mathbb{C} \setminus A$ is open.

Can you see what particular topology this definition allows us to equip $\mathbb{C}$ with?

2.3 Maximum and Supremum

2.3.1 The existence

Definition 2.3.1: Bounded Sets, Maxima and Minima

Let $X \subseteq \mathbb{R}$ be a subset of the real numbers.

- X is **bounded from above** if there exists $s \in \mathbb{R}$ such that $x \leq s$ for all $x \in X$. Any such $s \in \mathbb{R}$ is called an **upper bound** of X . An upper bound that is itself contained in X is called a **maximum**. We write

$$s = \max(X)$$

if such a maximum of X exists and is equal to s .

- The terms **bounded from below**, **lower bound**, and **minimum** are defined analogously.
- $X \subseteq \mathbb{R}$ is called **bounded** if it is bounded from above and from below.

Note that if a maximum (or a minimum) exists, then it is unique. This is why we say *the* maximum (or minimum). With a bit more of generality, we can give the following definition.

Definition 2.3.2: Supremum

Let $X \subseteq \mathbb{R}$ be a subset and let $A := \{a \in \mathbb{R} \text{ such that } x \leq a \text{ for all } x \in X\}$ be the set of all upper bounds of X . If the minimum $s = \min(A)$ exists, then we call it the **supremum** of x , and write

$$s = \sup(X).$$

In words, the supremum (assuming that it exists) of X is the smallest upper bound of X .

To see why this definition generalizes the previous one, consider a set $X \subseteq \mathbb{R}$, and suppose that $s = \max(X)$. In this case, we have $x \leq s$ for all $x \in X$. Also, since s is in X itself, for every upper bound a of X , we have $a \geq s$, and therefore $s = \sup(X)$. Hence, if the maximum exists, then it coincides with the supremum.

We here want to stress the fact that, again, we defined an object (namely, the supremum), but we did not address the question of its existence. The next result comes in our help.

Theorem 2.3.1: Existence of the Supremum

Let $X \subset \mathbb{R}$ be a nonempty subset bounded from above. Then the supremum of X exists.

It is interesting to see how the existence of the supremum and its value interact with the usual set operations. Before stating the next proposition, consider two non-empty sets $X, Y \subseteq \mathbb{R}$ both bounded from above. We write

$$X + Y := \{x + y \text{ such that } x \in X, y \in Y\}, \text{ and } XY := \{xy \text{ such that } x \in X, y \in Y\}.$$

Proposition 2.3.1: Supremum vs Set Operations

Let X, Y be nonempty subsets of $\mathbb{R}$ both bounded from above. Then, the sets $X \cup Y$, $X \cap Y$, and $X + Y$ are bounded from above. Also, if $x \geq 0$ and $y \geq 0$ for all $x \in X$ and $y \in Y$, then XY is bounded from above. And we have

1. $\sup(X \cup Y) = \max\{\sup(X), \sup(Y)\}$.
2. If $X \cap Y$ is again nonempty, then $\sup(X \cap Y) \leq \min\{\sup(X), \sup(Y)\}$.
3. $\sup(X + Y) = \sup(X) + \sup(Y)$.
4. If $x, y \geq 0$ for all $x \in X$ and $y \in Y$, then $\sup(XY) = \sup(X) \sup(Y)$.

We don't dwell on details here, but a symmetric discussion applies to the concept of **infimum**. Try to get convinced by yourself, maybe by observing that, if the infimum of a nonempty subset $X \subseteq \mathbb{R}$ bounded

from below is the largest lower bound of X , then one can show that

$$\inf(X) = -\sup(\{-x \text{ such that } x \in X\}).$$

2.3.2 Two-point Compactification

We have seen that the concepts of supremum and infimum already represent a generalization of the ones of, respectively, maximum and minimum. Nevertheless, we still encounter some sets that appear not to have an infimum or a supremum, while our intuition would suggest differently. To fix this problem, we introduce the **extended number line**, which is also called **two-point compactification** of $\mathbb{R}$, defined as

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{-\infty, \infty\}.$$

Definition 2.3.3: Indefinite Values

Let X be a subset of $\mathbb{R}$. If $X \subseteq \mathbb{R}$ is not bounded from above, then we define $\sup(X) = \infty$. If, instead, X is empty, we set $\sup(X) = -\infty$. In this context, we call ∞ and $-\infty$ **indefinite values**.

2.4 Consequences of Completeness

Again, $\mathbb{R}$ is an arbitrary set of real numbers. We now discuss further consequences of the completeness axiom.

2.4.1 The Archimedean Principle

Theorem 2.4.1: Archimedean Principle

For every $x \in \mathbb{R}$, there exists exactly one $n \in \mathbb{Z}$ with $n \leq x < n + 1$.

Usually, Archimedes' principle is stated in the following, weaker way: for every real number, there is an integer greater than it.

Definition 2.4.1: Integer and Fractional Parts

The **integer part** $\lfloor x \rfloor$ of a number $x \in \mathbb{R}$ is the integer $n \in \mathbb{Z}$ uniquely determined by the Archimedean principle with $n \leq x < n + 1$. The function $x \mapsto \lfloor x \rfloor$ is called **rounding function**. The **fractional part** of a real number x is $\{x\} := x - \lfloor x \rfloor$.

Being beyond our scope, we don't provide a formal definition here, but we still want to point out that, symmetrically, one can define the **ceiling** of x as the smallest integer that is greater than x . The ceiling of x is usually denoted by $\lceil x \rceil$.

Theorem 2.4.2: $\frac{1}{n}$ is arbitrarily small

For every $\epsilon > 0$ there exists an integer $n \geq 1$ such that $\frac{1}{n} < \epsilon$ holds.

Corollary 2.4.1: Density of $\mathbb{Q}$

For every two real numbers $a, b \in \mathbb{R}$, with $a < b$, there exists one $r \in \mathbb{Q}$ with $a < r < b$.

Stated differently, this corollary shows that $\mathbb{Q}$ intersects any open non-empty interval $I \subseteq \mathbb{R}$. A set with this property is called **dense** in $\mathbb{R}$, and thus $\mathbb{Q}$ is a dense (proper) subset in $\mathbb{R}$. This property is very important and constitutes the basis on which many proofs rely. In other words, it says that, for every fixed real number x , and any fixed level of error $\epsilon > 0$, it is possible to find some $q \in \mathbb{Q}$ such that $|x - q| < \epsilon$, i.e., q approximates the value of x within the level of precision ϵ .

It's not our intent to dwell on the details of this here, but we deemed it worth introducing the following question. The set $\mathbb{Q}$, as well as $\mathbb{N}$ and $\mathbb{Z}$, is a so-called **countable** set, while $\mathbb{R}$ is not. Intuitively, this means that the elements of $\mathbb{R}$ can't be listed in any way, while it's possible to list natural, integer, and rational numbers. In some sense, this also means that $\mathbb{R}$ is somehow "wilder" than the other three sets. What's then nice about this last corollary is that, despite the wildness of $\mathbb{R}$, it's still possible to approximate any real number within any level of precision by using numbers that belong to a more handy set, $\mathbb{Q}$.

2.5 Sequences of Real Numbers

2.5.1 Convergence of Sequences

Definition 2.5.1: Sequences

A **sequence** in $\mathbb{R}$ is a function $a : \mathbb{N} \rightarrow \mathbb{R}$. The image $a(n)$ of $n \in \mathbb{N}$ is also written as a_n and is called the **n -th element** of a . Instead of $a : \mathbb{N} \rightarrow \mathbb{R}$, we also write $(a_n)_{n \in \mathbb{N}}$.

As we'll see, what's usually important about a sequence is how the sequence behaves for large n , so that the first elements of the sequence are not relevant for many of our purposes. This is why we need definitions like the following.

Definition 2.5.2: (Eventually) Constant Sequences

A sequence $(x_n)_{n \in \mathbb{N}}$ is called **constant** if $x_n = x_{n+1}$ for all $n \in \mathbb{N}$, and **eventually constant** if there exists $N \in \mathbb{N}$ such that $x_n = x_m$ for all $m, n \in \mathbb{N}$ with $m, n \geq N$.

Definition 2.5.3: Convergence for Sequences

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. We say that $(x_n)_{n \in \mathbb{N}}$ is **convergent** or **converges** if there exists $A \in \mathbb{R}$ with the following property:

$$\forall \epsilon, \text{ there exists } N \in \mathbb{N} \text{ such that } \forall n \in \mathbb{N}, \text{ we have: } n \geq N \implies |x_n - A| < \epsilon.$$

In this case, we write

$$\lim_{n \rightarrow \infty} x_n = A,$$

and we call A the **limit** of $(x_n)_{n \in \mathbb{N}}$.

Note that, a priori, it is not clear that a converging sequence has only one limit. The nice thing is that, indeed, this is true!

Lemma 2.5.1: Uniqueness of the Limit

A convergent sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ has exactly one limit.

Note also that we do need to assume that a sequence converges, if we want the uniqueness of the limit. Indeed, a sequence is in general not converging. Yet, it is possible, as we shall see soon, to weaken a bit the concept of limit in order to define a sort of limit points for whatever sequences. First, we introduce a new concept.

Definition 2.5.4: Subsequences

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. A **subsequence** of $(x_n)_{n \in \mathbb{N}}$ is a sequence of the form $(x_{n_k})_{k \in \mathbb{N}}$, where $(n_k)_{k \in \mathbb{N}}$ is an increasing sequence of nonnegative integers (i.e., $n_{k+1} > n_k$ for all $k \in \mathbb{N}$).

Lemma 2.5.2: Subsequences of Convergent Sequences are Convergent

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$ converging to $A \in \mathbb{R}$. Then, each subsequence of $(x_n)_{n \in \mathbb{N}}$ also converges to A .

What's important to notice is that the convergence of a sequence is a sufficient condition for the convergence of one (or more) of their subsequences but not a necessary one. Can you provide an example of a non-converging sequence with a converging subsequence?

Definition 2.5.5: Accumulation Points of Sequences

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. A point $A \in \mathbb{R}$ is called an **accumulation point** of $(x_n)_{n \in \mathbb{N}}$ if for every $\epsilon > 0$ and any $N \in \mathbb{N}$ there exists a natural number $n \geq N$ such that $|x_n - A| < \epsilon$.

One may now think that, if A is an accumulation point of some sequence $(x_n)_{n \in \mathbb{N}}$, then it's also its limit. Yet, this is not true. Do you have a counterexample?

Proposition 2.5.1: Subsequences and Accumulation Points

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. An element $A \in \mathbb{R}$ is an accumulation point of $(x_n)_{n \in \mathbb{N}}$ if and only if there exists a convergent subsequence of $(x_n)_{n \in \mathbb{N}}$ with limit A .

Corollary 2.5.1: Accumulation Points have Infinite Elements Nearby

Let $A \in \mathbb{R}$ be an accumulation point of the sequence $(x_n)_{n \in \mathbb{N}}$. Then, for every $\epsilon > 0$, there are infinitely many elements of the sequence $(x_n)_{n \in \mathbb{N}}$ inside $(A - \epsilon, A + \epsilon)$.

Corollary 2.5.2: Accumulation Points of Convergent Sequences

A converging sequence has exactly one accumulation point, which coincides with its limit.

2.5.2 Addition, Multiplication, and Inequalities

Do you remember our discussion about algebraic structures? After encountering the ordered field of the real numbers, we are now presented with an example of a commutative ring. Given two sequences $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$, and a number $\alpha \in \mathbb{R}$, sums and scalar multiples of sequences are given by

$$(x_n)_{n \in \mathbb{N}} + (y_n)_{n \in \mathbb{N}} := (x_n + y_n)_{n \in \mathbb{N}}, \quad (2.2)$$

$$\alpha \cdot (x_n)_{n \in \mathbb{N}} := (\alpha x_n)_{n \in \mathbb{N}}. \quad (2.3)$$

Moreover, we can also define the product of the two sequences as

$$(x_n)_{n \in \mathbb{N}} \cdot (y_n)_{n \in \mathbb{N}} := (x_n y_n)_{n \in \mathbb{N}}.$$

With the addition and multiplication defined above, the set of sequences forms a commutative ring, where the zero element is the constant sequence $(0)_{n \in \mathbb{N}}$, and the neutral element for multiplication is $(1)_{n \in \mathbb{N}}$. The next result shows that, in some sense, the limits and the operations for sequences interact nicely.

Proposition 2.5.2: Limits and Operations

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences converging to $A \in \mathbb{R}$ and $B \in \mathbb{R}$ respectively. We then have:

1. The sequence $(x_n + y_n)_{n \in \mathbb{N}}$ converges to $A + B$.
2. The sequence $(x_n y_n)_{n \in \mathbb{N}}$ converges to AB , and the sequence $(\alpha x_n)_{n \in \mathbb{N}}$ converges to αA .
3. Suppose $x_n \neq 0$ for all $n \in \mathbb{N}$ and $A \neq 0$. Then the sequence $(x_n^{-1})_{n \in \mathbb{N}}$ converges to A^{-1} .

In 3., we could in fact weaken the assumption on $(x_n)_{n \in \mathbb{N}}$. Do you see how? Do you remember what we said about the behavior of a sequence for large n ?

Proposition 2.5.3: Limits and Inequalities

Let $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ be sequences of real numbers with limits $A \in \mathbb{R}$ and $B \in \mathbb{R}$ respectively. Then, we have:

1. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for all $n \geq N$.
2. Assume that $x_n \leq y_n$ for all $n \geq N$ for some $N \in \mathbb{N}$. Then $A \leq B$.

Lemma 2.5.3: Sandwich Lemma

Let $(x_n)_{n \in \mathbb{N}}$, $(y_n)_{n \in \mathbb{N}}$, and $(z_n)_{n \in \mathbb{N}}$ be sequences of real numbers such that for some $N \in \mathbb{N}$ the inequalities $x_n \leq y_n \leq z_n$ hold for all $n \geq N$. Suppose that $(x_n)_{n \in \mathbb{N}}$ and $(z_n)_{n \in \mathbb{N}}$ are convergent and have the same limit. Then the sequence $(y_n)_{n \in \mathbb{N}}$ is also convergent, and it holds that

$$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n = \lim_{n \rightarrow \infty} z_n.$$

2.5.3 Bounded Sequences**Definition 2.5.6: Bounded sequences**

A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is called **bounded** if there exists a real number $M \geq 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Lemma 2.5.4: Convergent Sequences are Bounded

If a (real-valued) sequence is convergent, then it is also bounded.

It may be useful to look at the proof of this lemma in detail. It uses a very standard argument, that could turn out to be useful and will prove to be ubiquitous in your next studies. We refer to the following idea: given a sequence $(x_n)_{n \in \mathbb{N}}$ that converges, say, to $A \in \mathbb{R}$, we know that $|x_n - A|$ is uniformly bounded for n sufficiently large. Therefore, we can set an “error” $\epsilon > 0$, and then choose $N \in \mathbb{N}$ such that $A - \epsilon \leq x_n \leq A + \epsilon$ for all $n \geq N$. This shows that $(x_n)_{n \geq N}$ is bounded (for example, by $|A| + \epsilon$). Now, we can consider the set $\{x_0, \dots, x_{N-1}\}$, and observe that, since it is finite, it admits a maximum, say M . Hence, the sequence $(x_n)_{n \in \mathbb{N}}$ must be bounded by $\max(M, |A| + \epsilon)$.

Definition 2.5.7: Monotone Sequence

A sequence $(x_n)_{n \in \mathbb{N}}$ is called:

- **(monotonically) increasing** if for all $m, n \in \mathbb{N}$ with $m > n$, we have $x_m \geq x_n$;
- **(strictly) increasing** if for all $m, n \in \mathbb{N}$ with $m > n$, we have $x_m > x_n$.

Similarly, one defines **(monotonically) decreasing** and **(strictly) decreasing** sequences.

If a sequence is increasing or decreasing, we call it **monotone**. If it is strictly increasing or strictly decreasing, we call it **strictly monotone**.

Theorem 2.5.1: Convergence for Monotone Sequences

A monotone sequence of real numbers converges if and only if it is bounded.
 If the sequence $(x_n)_{n \in \mathbb{N}}$ is monotonically increasing, then

$$\lim_{n \rightarrow \infty} x_n = \sup\{x_n | n \in \mathbb{N}\}.$$

If the sequence $(x_n)_{n \in \mathbb{N}}$ is monotonically decreasing, then

$$\lim_{n \rightarrow \infty} x_n = \inf\{x_n | n \in \mathbb{N}\}.$$

As suggested above, a sequence is not necessarily convergent. Nevertheless, it could be useful to have a weaker notion than the one of limit, so that for any sequence, one may still talk of some sort of convergence. At the moment, we are still unable to appreciate such generality, but after introducing the improper limits for a sequence, this comment will prove to be clearer.

Definition 2.5.8: Superior and Inferior Limits

Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence of real numbers. The real numbers defined by

$$\limsup_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} (\sup\{x_k | k \geq n\})$$

and

$$\liminf_{n \rightarrow \infty} x_n := \lim_{n \rightarrow \infty} (\inf\{x_k | k \geq n\})$$

are called **superior limit**, respectively **inferior limit** of $(x_n)_{n \in \mathbb{N}}$.

An immediate corollary of the definition is that the inferior limit of a sequence is no greater than its superior limit. The next lemma says something interesting about the case in which they coincide.

Lemma 2.5.5: Convergence vs Superior and Inferior Limits

A bounded sequence in $\mathbb{R}$ converges if and only if its superior limit and inferior limit coincide.

Theorem 2.5.2: Superior and Inferior Limits are Accumulation Points

Let $(x_n)_{n \in \mathbb{N}}$ be a bounded sequence whose superior limit is $A \in \mathbb{R}$. Then A is an accumulation point and, for every $\epsilon > 0$, the following hold:

- (1) there are only finitely many elements satisfying $x_n \geq A + \epsilon$;
- (2) infinitely many elements satisfy $A - \epsilon < x_n < A + \epsilon$.

An analogous statement holds for the inferior limit.

Corollary 2.5.3: Bounded Sequences have Convergent Subsequences

Every bounded sequence of real numbers has an accumulation point and a convergent subsequence.

2.5.4 Cauchy Sequences**Definition 2.5.9: Cauchy Sequences**

A sequence $(x_n)_{n \in \mathbb{N}}$ in $\mathbb{R}$ is called a **Cauchy sequence** if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - x_m| < \epsilon$ for all $m, n \in \mathbb{N}$ with $m, n \geq N$.

Intuitively, a Cauchy sequence is such that the elements of the sequence get closer and closer to each other. Mmh, should not a Cauchy sequence have also a limit, then?

Lemma 2.5.6: Boundedness of Cauchy Sequences

A Cauchy sequence is bounded.

Theorem 2.5.3: Convergence vs Cauchy Sequences

A sequence of real numbers converges if and only if it is a Cauchy sequence.

As a side note, we want to mark the importance of this result. In mathematical analysis, a metric space (which has not been precisely defined here) is called **complete** if every Cauchy sequence of points in that space is convergent. This definition is non-trivial, in the sense that there exist metric spaces that are not complete. Being $\mathbb{R}$ a metric space, with $|\cdot|$ defining its distance, it then turns out to be complete, thanks to the previous theorem. This makes us appreciate another peculiar characteristic of the set of real numbers. If you want to find an example of a non-complete metric space, you could consider the set of rational numbers, $\mathbb{Q}$. Do you see any Cauchy sequence that is not convergent?

2.5.5 Improper limits**Definition 2.5.10: Improper Limits**

Let $(x_n)_{n \in \mathbb{N}}$ be a sequence of real numbers. We say $(x_n)_{n \in \mathbb{N}}$ **diverges to** ∞ , and we write

$$\lim_{n \rightarrow \infty} x_n = \infty$$

if for every $M > 0$ there exists $N \in \mathbb{N}$ such that $x_n > M$ for all $n \geq N$.

Similarly, we say that $(x_n)_{n \in \mathbb{N}}$ **diverges to** $-\infty$ if for every real number $M > 0$ there exists $N \in \mathbb{N}$ such that $x_n < -M$ for all $n \geq N$.

In both cases, we speak of an **improper limit**.

With this new notion, we can now look back at the definition of superior and inferior limits, and see that these two objects can always be defined, should they be proper or improper.

2.6 Sequences of Complex Numbers

Definition 2.6.1: Sequences of Complex Numbers

A sequence of complex numbers $(z_n)_{n \in \mathbb{N}} = (x_n + iy_n)_{n \in \mathbb{N}}$ is convergent with limit $A + iB \in \mathbb{C}$, with $A, B \in \mathbb{R}$ if the two sequences of real numbers $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are convergent, with limits A and B respectively.

We say that $(z_n)_{n \in \mathbb{N}}$ diverges to ∞ if the sequence $(|z_n|)_{n \in \mathbb{N}}$ diverges to ∞ .

Chapter 3

Functions of one Real Variable

The central concept of this chapter is the one of continuity.

3.1 Real-valued Functions

3.1.1 Boundedness and Monotonicity

A **real-valued** function is any function which takes values on $\mathbb{R}$. We will assume that the domain of the functions we consider is a non-empty subset of $\mathbb{R}$. Given two real-valued functions having the same domain, one can define their sum, their product, and the scalar multiplication, in the same way we defined these same operations for sequences. Let $f : D \rightarrow \mathbb{R}$ be some real-valued function, with domain D . We call $x \in D$ a **zero** of f if $f(x) = 0$. The **zero set** of f is defined as $\{x \in D | f(x) = 0\}$. Finally, given any two functions f_1, f_2 in $\mathcal{F}(D) := \{\text{real-valued functions defined on } D\}$, we say

$$\begin{aligned} f_1 \leq f_2 &\Leftrightarrow f_1(x) \leq f_2(x) \text{ for all } x \in D, \\ f_1 \geq f_2 &\Leftrightarrow f_1(x) \geq f_2(x) \text{ for all } x \in D. \end{aligned}$$

Definition 3.1.1: Bounded Functions

Let $D \subseteq \mathbb{R}$ be a non-empty set and $f : D \rightarrow \mathbb{R}$ a function. We say that f is **bounded from above** if there exists some $M > 0$ such that

$$f(x) \leq M \text{ for all } x \in D.$$

We say that f is **bounded from below** if there exists some $M > 0$ such that

$$f(x) \geq -M \text{ for all } x \in D.$$

Finally, we say that f is **bounded** if it is bounded both from above and below.

Definition 3.1.2: Monotone Functions

Let $D \subset \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$. The function f is called

1. **(monotonically) increasing** if for all $x, y \in D$, we have

$$x < y \implies f(x) \leq f(y);$$

2. **strictly (monotonically) increasing** if for all $x, y \in D$, we have

$$x < y \implies f(x) < f(y).$$

Symmetrically, one defines **(monotonically) decreasing** and **strictly (monotonically) decreasing** functions.

We call $f : D \rightarrow \mathbb{R}$ **monotone** if it is monotonically increasing or monotonically decreasing, and **strictly monotone** if it is strictly monotonically increasing or strictly monotonically decreasing.

3.1.2 Continuity

The concept of continuity is extremely important and it is the central one of this section. We will see that a function is continuous if, roughly speaking, it is possible to draw down its graph without ever detaching the pen from the sheet. Nevertheless, there are subtleties that our intuition may not be able to detect by looking at this simple (and imprecise) characterization. Let's then deepen our comprehension by looking at the formal definition.

Definition 3.1.3: Continuous Functions

Let $D \subseteq \mathbb{R}$ and let $f : D \rightarrow \mathbb{R}$ be a function. We say that f is **continuous at a point** $x_0 \in D$ if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$\forall x \in D, |x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon.$$

The function f is said to be **continuous** in D if it is continuous at every point of D .

The next result shows that, in some sense, continuous functions behave well when combined together.

Lemma 3.1.1: Combination of Continuous Functions

Let $D \subseteq \mathbb{R}$, and let $f_1, f_2 : D \rightarrow \mathbb{R}$ be functions both continuous at $x_0 \in D$. Then also the functions $f_1 + f_2$, $f_1 f_2$, and αf_1 for $\alpha \in \mathbb{R}$ are continuous at x_0 .

Not only do the addition, and the multiplication behave well for continuous functions, but also the composition does so. Let's see.

Proposition 3.1.1: Composition on Continuous Functions

Let D_1, D_2 be two (non-empty) subsets of $\mathbb{R}$, and let $x_0 \in D_1$. Let $f_1 : D_1 \rightarrow D_2$ be a continuous function at x_0 and let $g : D_2 \rightarrow \mathbb{R}$ be a continuous function at $f_1(x_0)$. Then the function $g \circ f_1 : D_1 \rightarrow \mathbb{R}$ is continuous at x_0 . In particular, the continuity is preserved by the composition.

The next theorem provides a different characterization of the concept of continuity, namely by the so-called sequential continuity.

Theorem 3.1.1: Continuity = Sequential Continuity

Let $D \subseteq \mathbb{R}$ be a subset, $f : D \rightarrow \mathbb{R}$ be a function, and $\bar{x} \in D$. The function f is continuous at $\bar{x}$ if and only if, however taken a sequence $(x_n)_{n \in \mathbb{N}} \subset D$ converging to $\bar{x}$, the sequence $(f(x_n))_{n \in \mathbb{N}}$ converges to $f(\bar{x})$.

3.2 Continuous Functions and their Properties

Now that we know what a continuous function is, we can start appreciating the power of this property by presenting some first results related to the continuity property.

Theorem 3.2.1: Intermediate Value Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function with $f(a) \leq f(b)$. Then, for every real number c with $f(a) \leq c \leq f(b)$, there exists $\bar{x} \in [a, b]$ such that $f(\bar{x}) = c$.

Recall what we said about “not detaching the pen from the sheet”. This theorem should then sound intuitive, shouldn’t it? In fact, it tells us that if a continuous function take on the values $f(a) < f(b)$, then it must take on also all the values c between $f(a)$ and $f(b)$.

3.2.1 Inverse Function Theorem

Theorem 3.2.2: Inverse Function Theorem

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \rightarrow \mathbb{R}$ be a continuous strictly monotone function. Then $f(I) \subseteq \mathbb{R}$ is again an interval, and the mapping $f : I \rightarrow f(I)$ has a continuous strictly monotone inverse function $f^{-1} : f(I) \rightarrow I$.

It is important to note that it is not the function $f : I \rightarrow \mathbb{R}$ that admits an inverse, but $f : I \rightarrow f(I)$. The latter is in general not surjective, and thus it is not sensible to speak of an inverse of it.

3.3 Continuous Functions on Compact Intervals

Before stating the results of this section, we need to introduce the notion of compactness. This is a central one in topology, and it can be defined in a context far more general than ours. If you look at the definition of a compact set, you will probably feel a bit confused. Nevertheless, as it often happens, a very obscure object assumes a way more handy aspect when in the context of real numbers. A subset of $\mathbb{R}$ is called **compact** if it is both bounded and closed.

3.3.1 Boundedness and Extrema

Lemma 3.3.1: Compactness

Let $[a, b]$ be a compact interval, and $(x_n)_{n \in \mathbb{N}}$ be a sequence contained in $[a, b]$. Then there is a subsequence $(x_{n_k})_{k \in \mathbb{N}}$ such that $\lim_{k \rightarrow \infty} x_{n_k} = \bar{x} \in [a, b]$.

Theorem 3.3.1: Boundedness

Let $[a, b]$ be a compact interval, and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then f is bounded.

Note that all the conditions we require in the previous theorem are necessary to guarantee the boundedness of f . To get convinced, try to remove one condition at a time, and find an example of a non-continuous f .

Definition 3.3.1: Extreme Values

Let $D \subseteq \mathbb{R}$ and $f : D \rightarrow \mathbb{R}$ be a real-valued function on D . We say that the function f takes its **maximal value** at a point $x_0 \in D$ if $f(x) \leq f(x_0)$ for all $x \in D$. We then call $f(x_0)$ the **maximum** of f (in D). Similarly, one can define the **minimal value** and the **minimum**. Maxima and minima are summarily called **extreme values** or **extrema**.

If the maximum and the minimum of a function are, when they exist, unique, this is not necessarily the case for the maximal and the minimal values. Do you see it? You can pick up any constant function, and see that, according to the definition, any x in its domain is both a maximal and minimal value.

Theorem 3.3.2: Extreme Value Theorem

Let $[a, b]$ be a compact interval, and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function. Then f attains both its maximum and minimum.

3.3.2 Uniform Continuity

Definition 3.3.2: Uniform Continuity

Let $D \subseteq \mathbb{R}$. A function $f : D \rightarrow \mathbb{R}$ is called **uniform continuous** if for all $\epsilon > 0$ there exists some $\delta > 0$ such that the implication

$$|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$$

holds for all $x, y \in D$.

This definition is in its fashion similar to the one of continuity. Yet, we have in addition the uniformity condition, which amounts to letting both x and y be arbitrary elements of the domain.

Try to reflect on the case of $f(x) = x^2$ as a function on $\mathbb{R}$ first and on $[0, 1]$ next, and, after proving that it is uniformly continuous in the latter case but not in the former, investigate the reason why the two setup lead to different conclusions.

Theorem 3.3.3: Compact Intervals and Uniform Continuity

Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function, with $a < b$. Then f is also uniformly continuous.

What does happen if $[a, b]$ is replaced by $(0, 1)$? Does the result keep holding? To find the answer, try to look at the function $f(x) = \frac{1}{x}$, which is clearly continuous on $(0, 1)$.

Definition 3.3.3: Lipschitz Continuity

Let $D \subseteq \mathbb{R}$ be a subset. A function $f : D \rightarrow \mathbb{R}$ is called **Lipschitz continuous** if there exists some $L \geq 0$ such that

$$|f(x) - f(y)| \leq L|x - y| \text{ for all } x, y \in D.$$

What can you say about the relationship between the three types of continuity introduced so far? A good exercise is the one of showing what type of continuity implies what other. You should see that

$$\text{Lipschitz continuity} \implies \text{Uniform continuity} \implies \text{Continuity}.$$

3.4 Limits of Functions

We consider a function $f : D \rightarrow \mathbb{R}$ on a subset $D \subseteq \mathbb{R}$ and want to define limits of f when $x \in D$ tends to a point $x_0 \in \mathbb{R}$. By doing this, we will be able to investigate the behavior of a function at the boundaries of its domain, and, more generally, in the vicinity of all its so-called accumulation points.

3.4.1 Limit in the Vicinity of a Point

Let $D \subseteq \mathbb{R}$ be a subset, and $x_0 \in \mathbb{R}$ such that

$$D \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$$

for all δ . Such a point is called an **accumulation point**.

Definition 3.4.1: Limit of a Function

Let $f : D \rightarrow \mathbb{R}$, and assume that $D \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for every $\delta > 0$.

A real number $L \in \mathbb{R}$ is called **limit** of $f(x)$ as $x \rightarrow x_0$ if for every $\epsilon > 0$, there exists δ such that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies |f(x) - L| < \epsilon.$$

In general, the limit of $f(x)$ as $x \rightarrow x_0$ may not exist. But if it exists, then it is uniquely determined (note that this fact is not trivial and has to be proved). Therefore, from now on, we speak of *the* limit and write

$$\lim_{x \rightarrow x_0} f(x) = L$$

if the limit of $f(x)$ as $x \rightarrow x_0$ exists and is equal to L .

Suppose that $x_0 \in D$ is an accumulation point of $D \setminus \{x_0\}$. Let $f : D \rightarrow \mathbb{R}$ be a function and consider its restriction $f|_{D \setminus \{x_0\}} : D \setminus \{x_0\} \rightarrow \mathbb{R}$ to the set $D \setminus \{x_0\}$, and further assume that this function is continuous. f may be discontinuous at the point x_0 , but the limiting value at x_0 of its restriction may nevertheless exist. In this case, we call x_0 a **removable discontinuity**. In such a case, it is possible to *extend with continuity* the restriction to the set D by defining a new function $\tilde{f}$, which coincides with f on $D \setminus \{x_0\}$ and takes on the value of the limit of $f|_{D \setminus \{x_0\}}$ at x_0 . We call this new function the **continuous extension** of $f|_{D \setminus \{x_0\}}$ on D .

Lemma 3.4.1: Limit vs Sequences

Let $f : D \rightarrow \mathbb{R}$. Then $L = \lim_{x \rightarrow \bar{x}} f(x)$ if and only if, for every sequence $(x_n)_{n \in \mathbb{N}} \subset D$ converging to $\bar{x}$, $\lim_{n \in \mathbb{N}} f(x_n) = L$ also holds.

Proposition 3.4.1: Limit and Composition

Let $E \subset \mathbb{R}$, and let $f : D \rightarrow E$ be such that the limit $\bar{y} = \lim_{x \rightarrow \bar{x}} f(x)$ exists and belongs to E . Let $g : E \rightarrow \mathbb{R}$ be continuous at $\bar{y}$. Then $\lim_{x \rightarrow \bar{x}} g(f(x)) = g(\bar{y})$ holds.

As done with the sequences, we can introduce improper limits.

Definition 3.4.2: Divergence for Sequences

Let $f : D \rightarrow \mathbb{R}$, and assume that $D \cap (x_0 - \delta, x_0 + \delta) \neq \emptyset$ for every $\delta > 0$.

We say that f **diverges** to ∞ as $x \rightarrow x_0$, and write $\lim_{x \rightarrow x_0} f(x) = \infty$, if for every real number M there exists δ with the property that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies f(x) \geq M \text{ for all } x \in D.$$

A symmetric definition is given for the limit $-\infty$.

3.4.2 One-sided Limits

The concept of limit allows us to study the behavior of (the graph of) a function in the vicinity of a certain point of its domain. In the definition, by “vicinity”, we refer to points both on the left-hand side and on the right-hand side of the point under consideration. In practice, this may be too strict: in fact, we may be interested in studying the behavior of a function $f : I \rightarrow \mathbb{R}$ in the vicinity of the extreme points of I , where I is some interval. In this case, the function is defined only on one side of the point at which we intend to evaluate the limit. To deal with these more general cases, we introduce a new definition.

Definition 3.4.3: One-sided Limit

Let $f : D \rightarrow \mathbb{R}$, and assume that $D \cap [x_0, x_0 + \delta)$ is non-empty for every $\delta > 0$.

A real number L is called **limit from the right** of f at x_0 if for every $\epsilon > 0$ there exists some $\delta > 0$ such that

$$x \in D \cap [x_0, x_0 + \delta) \implies |f(x) - L| < \epsilon.$$

Similarly, one defines the **limit from the left**.

When the limit from the right of f exists at x_0 (and its value is L), we use the notation

$$L = \lim_{x \rightarrow x_0^+} f(x).$$

Definition 3.4.4: Limits at Infinity

Let $f : D \rightarrow \mathbb{R}$, and assume $D \cap (R, \infty) \neq \emptyset$ for every $R > 0$.

A real number L is called **limit** of f as $x \rightarrow \infty$ if for every $\epsilon > 0$ there exists some $R > 0$ such that

$$x \in D \cap (R, \infty) \implies |f(x) - L| < \epsilon.$$

We say that f **diverges** to ∞ as $x \rightarrow \infty$ if for every $M > 0$ there exists $R > 0$ such that

$$x \in D \cap (R, \infty) \implies f(x) \geq M.$$

By using the concept of one-sided limit introduced above, we can make sense of the concept of one-sided continuity.

Definition 3.4.5: One-sided Continuity and Jumps

Let $f : D \rightarrow \mathbb{R}$, and let $x_0 \in D$. If the limit from the right

$$\lim_{x \rightarrow x_0^+} f(x)$$

exists and it is equal to $f(x_0)$, then we say that f is **continuous from the right** at x_0 . Similarly, we define **continuous from the left**.

We call $x_0 \in D$ a **jump point** if the one-sided limits

$$\lim_{x \rightarrow x_0^+} f(x) \text{ and } \lim_{x \rightarrow x_0^-} f(x)$$

both exist but are different.

3.4.3 Landau Notation

As we will soon discover, it is often important to understand the *asymptotic* behavior of a function in the vicinity of a point or at infinity, that is, its behavior as $x \rightarrow x_0$, for some x_0 (x_0 can also be ∞). Thus, it may be useful to have a compact notation that allows us to express the relationships between two functions in terms of their asymptotic behavior. This is what the Landau Notation provides us with.

Definition 3.4.6: Landau Big-O

Let $f, g : D \rightarrow \mathbb{R}$. We write

$$f(x) = O(g(x)) \text{ as } x \rightarrow x_0$$

if there exist $M > 0$ and $\delta > 0$ such that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies |f(x)| \leq M|g(x)|.$$

Then we say that f is a **big-O** of g as $x \rightarrow x_0$.

Is this relationship symmetric? Or, in other words, if you know that f is a big-O of g , can you also say that g is a big-O of f ?

In general, the answer is negative (try to provide some examples). When this happens, though, we may say that the two functions are of the same order in a neighbor of x_0 .

Definition 3.4.7: Landau Big-O at Infinity

Let $f, g : D \rightarrow \mathbb{R}$. We write

$$f(x) = O(g(x)) \text{ as } x \rightarrow \infty$$

if there exist $M > 0$ and $R > 0$ such that

$$x \in D \cap (R, \infty) \implies |f(x)| \leq M|g(x)|.$$

Note that the advantage of this notation is that we do not need to introduce explicitly the upper bound M . If we are not particularly interested in this constant, then we can concentrate on the essentials in calculations.

Let's now introduce the little-o notation. Note that this is not a counterpart of the one introduced above, but rather a stronger condition. It allows us to express compactly the fact that a function is asymptotically negligible with respect of another.

Definition 3.4.8: Little-o

Let $f, g : D \rightarrow \mathbb{R}$. We write

$$f(x) = o(g(x)) \text{ as } x \rightarrow x_0$$

if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$x \in D \cap (x_0 - \delta, x_0 + \delta) \implies |f(x)| \leq \epsilon|g(x)|.$$

Then we say that f is a **little-o** of g as $x \rightarrow x_0$.

if $g(x) \neq 0$ for all x in a neighborhood of x_0 , then $f(x) = o(g(x))$ as $x \rightarrow x_0$ is equivalent to

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

Definition 3.4.9: Landau Little-o at Infinity

Let $f, g : D \rightarrow \mathbb{R}$. We write

$$f(x) = o(g(x)) \text{ as } x \rightarrow \infty$$

if for every $\epsilon > 0$ there exists $R > 0$ such that

$$x \in D \cap (R, \infty) \implies |f(x)| \leq \epsilon|g(x)|.$$

Can you show that the relationship of “being a little-o” implies the one of “being a big-O”?

3.5 Sequences of Functions

As we did with the set of real numbers, we can consider a set X , possibly uncountable, and choose countably infinitely many elements of X , and index them by using the non-negative integers, $\mathbb{N}$. By doing so, we will end up constructing a sequence $(x_n)_{n \in \mathbb{N}}$ of elements of X . In this section, we consider as X the set of all the real-valued functions having the same domain $D \subseteq \mathbb{R}$, and thus sequences of functions.

Definition 3.5.1: Sequences of Functions

A **sequence** of real-valued functions on a subset $D \subseteq \mathbb{R}$ is a family of functions $f_n : D \rightarrow \mathbb{R}$ indexed by $\mathbb{N}$. The function f_n is called the **n -th element** of the sequence. One often writes $(f_n)_{n \in \mathbb{N}}$ to refer to the sequence of the f_n .

Again, we proceed similarly to how we did with sequences of real numbers, namely we define a notion (actually, two) of convergence for a sequence of functions.

Definition 3.5.2: Pointwise Convergence

Let $D \subseteq \mathbb{R}$, let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions $f_n : D \rightarrow \mathbb{R}$, and let $f : D \rightarrow \mathbb{R}$ be a function. We say that the sequence $(f_n)_{n \in \mathbb{N}}$ **converges pointwise** to f if for every $x \in D$ the sequence $(f_n(x))_{n \in \mathbb{N}}$ converges to $f(x)$. In this case, we call f the **pointwise limit** of $(f_n)_{n \in \mathbb{N}}$.

An interesting problem is the one of determining whether, upon assuming that the functions f_n all have some property, the pointwise limit f shares the same property. Unfortunately, this sort of “transference” usually doesn’t hold. For example, the continuity of the f_n does not ensure the continuity of f . This fact suggests that the pointwise convergence may be a bit too weak. No panic, then, we can simply require something stronger.

Definition 3.5.3: uniform Convergence

Let $D \subseteq \mathbb{R}$, let $(f_n)_{n \in \mathbb{N}}$ be a sequence of functions $f_n : D \rightarrow \mathbb{R}$, and let $f : D \rightarrow \mathbb{R}$ be a function. We say that the sequence $(f_n)_{n \in \mathbb{N}}$ **converges uniformly** to f if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and all $x \in D$ we have

$$|f_n(x) - f(x)| < \epsilon.$$

It may prove to be useful to reflect on the geometric meaning of the uniform convergence. Let’s have a look at the interpretation in terms of the graph of the functions.

Theorem 3.5.1: Uniform Convergence Preserves Continuity

Let $(f_n)_{n \in \mathbb{N}}$ be a sequence of continuous functions $f_n : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$, and assume that the sequence converges uniformly to some function $f : D \rightarrow \mathbb{R}$. Then f is continuous.

Chapter 4

Sereies and Power Series

We dedicate this chapter to the study of the so-called series, i.e., infinite sums. We will find ourselves dealing with objects that we already know, namely sequences of real numbers. A series is, in fact, nothing but a specific kind of sequence.

4.1 Series of Real Numbers

Definition 4.1.1: Convergent Series

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers, and let $A \in \mathbb{R}$. We say that the series $\sum_{k=0}^{\infty} a_k$ converges to A if

$$A = \lim_{n \rightarrow \infty} s_n,$$

where $(s_n = \sum_{k=0}^n a_k)_{n \in \mathbb{N}}$ is the sequence of the **partial sums** associated with $(a_n)_{n \in \mathbb{N}}$.

We call a_n the **n -th element** or the **n -th summand** of the series. We call the series $\sum_{k=0}^{\infty} a_k$ **convergent** if $(s_n)_{n \in \mathbb{N}}$ admits a limit, in which case we call it the **value of the series**. Otherwise, the series is said to be **not convergent**.

If $(s_n)_{n \in \mathbb{N}}$ diverges to ∞ (resp., to $-\infty$), we call the series $\sum_{k=0}^{\infty} a_k$ **divergent** to ∞ (resp., $-\infty$).

As we will see, the study of the convergence of a series $\sum_{k=0}^{\infty} a_k$ can often be done by checking some properties of the sequence $(a_n)_{n \in \mathbb{N}}$, through the so-called convergence criteria for the series. Let's start with a necessary condition for the convergence of a sequence.

Proposition 4.1.1: Necessary Condition for Convergence

If the series $\sum_{k=0}^{\infty} a_k$ converges, then the sequence $(a_n)_{n \in \mathbb{N}}$ is a **null sequence**, i.e., it satisfies

$$\lim_{n \rightarrow \infty} a_n = 0.$$

Lemma 4.1.1: Convergent of the Tail

Let $\sum_{k=0}^{\infty} a_k$ be a series. For each $N \in \mathbb{N}$, the series $\sum_{k=N}^{\infty} a_k$ is convergent if and only if the series $\sum_{k=0}^{\infty} a_k$ is convergent. In this case, we have

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{N-1} a_k + \sum_{k=N}^{\infty} a_k.$$

The same statement holds even by replacing convergent with divergent.

We hope you remember we said that, often in the study of a sequence and its properties, what really counts is what happens for large n (or when $n \rightarrow \infty$). This previous lemma is a reformulation of this (not rigorous) observation. Indeed, if some tail of a series converges, then the series itself converges.

4.1.1 Series with Non-negative Elements

A special status among series is that of series with non-negative elements, in a sense made precise by the next result.

Proposition 4.1.2: Non-negative Series either Converge or Diverge

Given a series $\sum_{k=0}^{\infty} a_k$ with non-negative elements $a_k \geq 0$ for all $k \in \mathbb{N}$, the partial sums $s_n = \sum_{k=0}^n a_k$ form a monotonically increasing sequence. If the sequence $(s_n)_{n \in \mathbb{N}}$ is bounded, then the series $\sum_{k=0}^{\infty} a_k$ converges; otherwise, it diverges to ∞ .

In particular, this means that a series of non-negative elements must either converge or diverge. This is very reasonable: s_{n+1} is obtained from s_n by adding up a non-negative term, and thus the sequence of partial sums is monotonically non-decreasing. Therefore, it allows for only two options, that is, either it has a finite limit or it is not bounded, and so diverges. Clearly, if you take a series of both positive and negative elements, a new possibility pops out: the series can indeed be neither convergent nor divergent, but be oscillating.

What we do now is introduce some first criteria that provide us with useful rules aimed at determining whether a series converges or not.

Corollary 4.1.1: Majorant and Minorant Criterion

Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be two series such that $0 \leq a_k \leq b_k$ for all $k \in \mathbb{N}$. Then $0 \leq \sum_{k=0}^{\infty} a_k, \sum_{k=0}^{\infty} b_k$ holds, and, in particular, we have

$$\begin{aligned} \sum_{k=0}^{\infty} b_k \text{ convergent} &\implies \sum_{k=0}^{\infty} a_k \text{ convergent}, \\ \sum_{k=0}^{\infty} a_k \text{ divergent to } \infty &\implies \sum_{k=0}^{\infty} b_k \text{ divergent to } \infty. \end{aligned}$$

As a remark, we observe that this result holds even if the chain of inequalities $0 \leq a_n \leq b_n$ holds only for $n \geq N$ for some $N \in \mathbb{N}$. Again, this has to do with the fact that the overall behavior of a series is determined by the one of any tail.

Proposition 4.1.3: Cauchy Condensation Test

Let $(a_n)_{n \in \mathbb{N}}$ be a monotonically decreasing sequence of non-negative real numbers. Then

$$\sum_{k=0}^{\infty} a_k \text{ converges} \Leftrightarrow \sum_{k=0}^{\infty} 2^k a_{2^k} \text{ converges}.$$

4.1.2 Conditional Convergence

Definition 4.1.2: Absolute and Conditional Convergence

We say that a series $\sum_{k=0}^{\infty} a_k$ is **absolutely convergent** if the series $\sum_{k=0}^{\infty} |a_k|$ converges. The series $\sum_{k=0}^{\infty} a_k$ is **conditionally convergent** if it converges but is not absolutely convergent.

At first sight, one may think that conditionally convergent series still have a convergence property, that thus makes them “nice” objects. The next result, yet, casts doubt on this naive idea.

Theorem 4.1.1: Riemann's Rearrangement Theorem

Let $\sum_{k=0}^{\infty} a_k$ be a conditionally convergent series, and let $A \in \mathbb{R}$. Then, there exists a bijection $\phi : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$A = \sum_{n=0}^{\infty} a_{\phi(n)}.$$

In words, this means that, given a conditionally convergent series, one can rearrange the terms to obtain any possible limit. It is surprising, isn't it? One classic example of a conditionally convergent series is the following. Let $a_k := \frac{(-1)^k}{k}$, $k \in \mathbb{N}$, and consider the series $\sum_{k=0}^{\infty} a_k$. This series is called **alternating harmonic series**, and the series $\sum_{k=0}^{\infty} |a_k|$ is the so-called **harmonic series**.

4.1.3 Convergence Criteria of Leibniz and Cauchy**Definition 4.1.3: Alternating series**

Given a sequence $(a_n)_{n \in \mathbb{N}}$, we call the series $\sum_{k=0}^{\infty} (-1)^k a_k$ the corresponding **alternating series**.

Proposition 4.1.4: Leibniz Criterion

Let $(a_n)_{n \in \mathbb{N}}$ be a monotonically decreasing sequence of non-negative real numbers converging to zero. The alternating series $\sum_{k=0}^{\infty} (-1)^k a_k$ converges and

$$\sum_{k=0}^{2n+1} (-1)^k a_k \leq \sum_{k=0}^{\infty} (-1)^k a_k \leq \sum_{k=0}^{2n} (-1)^k a_k$$

holds for all $n \in \mathbb{N}$.

Theorem 4.1.2: Cauchy Criterion

The series $\sum_{k=0}^{\infty} a_k$ converges if and only if for every $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that, for $n \geq m \geq N$, we have

$$\left| \sum_{k=m+1}^n a_k \right| < \epsilon.$$

4.2 Absolute Convergence

In this section, we will look at absolutely convergent series and some convergence criteria. As in the previous discussion, unless otherwise specified, all sequences are meant to be of real numbers.

4.2.1 Criteria for Absolute Convergence

Proposition 4.2.1: Absolute Convergence implies Convergence

An absolutely convergent series $\sum_{k=0}^{\infty} a_k$ is also convergent and satisfies the generalized triangle inequality

$$\left| \sum_{k=0}^{\infty} a_k \right| \leq \sum_{k=0}^{\infty} |a_k|.$$

Corollary 4.2.1: Cauchy Root Criterion

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers, and let

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} \in \mathbb{R} \cup \{\infty\}.$$

Then, we have

$$\alpha < 1 \implies \sum_{k=0}^{\infty} a_k \text{ converges absolutely,}$$

and

$$\alpha > 1 \implies \sum_{k=0}^{\infty} a_k \text{ does not converge.}$$

Corollary 4.2.2: D'Alembert's quotient criterion

Let $(a_n)_{n \in \mathbb{N}}$ be a sequence of real numbers with $a_n \neq 0$ for all $n \in \mathbb{N}$, and assume that

$$\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|} \text{ exists.}$$

Let $\alpha \geq 0$ denote such a limit. Then, we have

$$\alpha < 1 \implies \sum_{k=0}^{\infty} a_k \text{ converges absolutely,}$$

and

$$\alpha > 1 \implies \sum_{k=0}^{\infty} a_k \text{ does not converge.}$$

As you can easily imagine, different criteria are usually not simultaneously applicable to the same series. Nevertheless, there are situations in which more than one criterium (among the ones presented here) can be used, and therefore one is spoiled for choice. Obviously, some criteria are preferable to others, simply because their conditions are easier to check. Let's look at the last two criteria that we have introduced. The condition on the sequence in D'Alembert's quotient criterion is usually way more handy than the one required by Cauchy's root criterium. Thus, one usually starts by using the latter. This is perfectly fine, but we need to note that there is a tradeoff here. If D'Alembert's criterion is easier to check, the limit $\lim_{n \rightarrow \infty} \frac{|a_{n+1}|}{|a_n|}$ does not need necessarily exist. As a consequence, we are not always able to apply this rule. On the other hand, the Cauchy criterion asks us to check the value of a superior limit, which is usually more difficult to compute, but yet always exists.

4.2.2 Reordering Series

Theorem 4.2.1: Rearrangement for Absolutely Convergent Series

Let $\sum_{k=0}^{\infty} a_k$ be an absolutely convergent series of real numbers. Let $\phi : \mathbb{N} \rightarrow \mathbb{N}$ be a bijection. Then the series $\sum_{k=0}^{\infty} a_{\phi(k)}$ is absolutely convergent, and

$$\sum_{k=0}^{\infty} a_k = \sum_{k=0}^{\infty} a_{\phi(k)}.$$

4.2.3 Product of Series

Theorem 4.2.2: Product Theorem

Let $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ be absolutely convergent series, and let $\alpha : \mathbb{N} \rightarrow \mathbb{N} \times \mathbb{N}$ be a bijection. Write $\alpha(n) = (\alpha_1(n), \alpha_2(n)) \in \mathbb{N} \times \mathbb{N}$ for each $n \in \mathbb{N}$. Then, we have

$$\left(\sum_{k=0}^{\infty} a_k \right) \left(\sum_{k=0}^{\infty} b_k \right) = \sum_{k=0}^{\infty} a_{\alpha_1(k)} b_{\alpha_2(k)},$$

and the series on the left converges absolutely.

Corollary 4.2.3: Cauchy Product

If $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ are absolutely convergent series, then

$$\left(\sum_{k=0}^{\infty} a_k \right) \left(\sum_{k=0}^{\infty} b_k \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^{\infty} a_{n-k} b_k \right),$$

where the series on the right-hand side is absolutely convergent.

4.3 Series of Complex Numbers

As in the case of sequences, to study series in $\mathbb{C}$, it is often sufficient to consider the corresponding series of real and imaginary parts in $\mathbb{R}$.

Definition 4.3.1: Series of Complex Numbers

Let $(z_n)_{n \in \mathbb{N}} = (x_n + iy_n)_{n \in \mathbb{N}}$ be a sequence of complex numbers, and let $Z = A + iB \in \mathbb{C}$. The series $(z_n)_{n \in \mathbb{N}}$ is convergent with limit Z if the two series $\sum_{k=0}^{\infty} x_k$ and $\sum_{k=0}^{\infty} y_k$ are convergent, with limits A and B , respectively.

We say that $\sum_{n=0}^{\infty} z_n$ converges absolutely if the series $\sum_{k=0}^{\infty} |z_k|$ converges.

4.4 Power Series

A nice class of series is that of the so-called power series, which we will define soon. The theory we develop here is applicable to both real-valued series and series with complex coefficients.

4.4.1 Radius of Convergence

Definition 4.4.1: Power Series

A **power series** with real coefficients is a series of the form

$$\sum_{n=0}^{\infty} a_n x^n,$$

where $(a_n)_{n \in \mathbb{N}}$ is a sequence in $\mathbb{R}$, and $x \in \mathbb{R}$. Here, we call x the **variable**, and the element a_n is called the **coefficient** of x^n .

Addition and multiplication of power series are given by simple formulas, where the one for the multiplication follows from a previous corollary. For the details, we refer to the lecture notes.

Definition 4.4.2: Radius of Convergence

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series. The **radius of convergence** of the series is the number $R \in \mathbb{R}_0^+$ or the symbol $R = \infty$, defined by

$$\rho := \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}, \quad \text{and} \quad R := \begin{cases} 0 & \text{if } \rho = \infty, \\ \rho^{-1} & \text{if } 0 < \rho < \infty, \\ \infty & \text{if } \rho = 0. \end{cases}$$

At this stage, the name “radius” may sound unjustified. The following section, which generalizes the next few results to the complex case, will put light on this terminology, which will then sound rather intuitive.

Theorem 4.4.1: Convergence of Power Series

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R \in (0, \infty) \cup \{\infty\}$. The series $\sum_{n=0}^{\infty} a_n x^n$ converges absolutely for all $x \in \mathbb{R}$ with $|x| < R$, and does not converge for all $x \in \mathbb{R}$ with $|x| > R$. In particular, for $x \in (-R, R)$, we can define the function $f(x) = \sum_{n=0}^{\infty} a_n x^n$.

Note that this theorem doesn’t say anything about the convergence of the series for $|x| = R$. This is neither due to laziness nor to negligence. In fact, we can’t say anything about the convergence for $|x| = R$ in such a generality.

Theorem 4.4.2: Continuity of Power Series

Let $\sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R \in \mathbb{R} \cup \{\infty\}$, and define the polynomials $f_n(x) := \sum_{k=0}^n a_k x^k$, $n \in \mathbb{N}$. For any $r < R$, the sequence $(f_n)_{n \in \mathbb{N}}$ converges uniformly to f on $(-r, r)$. In particular, the power series defines a continuous function $f : (-R, R) \rightarrow \mathbb{R}$.

Proposition 4.4.1: Radius of Convergence of Sum and Product

Let $R \geq 0$, and let $\sum_{n=0}^{\infty} a_n x^n$ and $\sum_{n=0}^{\infty} b_n x^n$ be both power series with radius of convergence at least R . Then, their sum and product also have radii of convergence at least R .

4.4.2 Complex Power Series

As stated above, we can define a power series with complex coefficients in the same fashion as how we introduced real power series. It’s then possible to extend (some of) the results we showed so far to this more

general setup.

For the sake of simplicity, we do not present them here, and we refer to the lecture notes.

Chapter 5

Differential Calculus

Here we are: finally, we can discuss differential calculus in one variable, a subject of fundamental importance for understanding functions on $\mathbb{R}$.

5.1 The Derivative

5.1.1 Definition and Geometrical Interpretation

In this section, we denote by $D \subseteq \mathbb{R}$ a non-empty set with no isolated points, i.e., every $x \in D$ is an accumulation point for $D \setminus \{x\}$.

Definition 5.1.1: Derivative

Let $f : D \rightarrow \mathbb{R}$ be a function, and let $x \in \mathbb{R}$. We say that f is **differentiable** at x_0 if the limit

$$f'(x_0) := \lim_{x \rightarrow x_0, x \neq x_0} \frac{f(x) - f(x_0)}{x - x_0} = \lim_{h \rightarrow 0, h \neq 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

exists. In this case, we call $f'(x_0)$ the **derivative** of f at x_0 . If f is differentiable at every point of D , then we say that f is **differentiable** on D , and we call the function $f' : D \rightarrow \mathbb{R}$ the **derivative** of f .

As a remark, we want to spend a couple of words on the usage of the word “differentiable”. In this context, where the functions are of one variable, we can speak of differentiability in relatively loose terms: the derivative exists, then the function is said to be differentiable. Later, you’ll see that this is not exactly the case for multivariate real-valued functions, with which one should be more careful.

By using the concepts of left- and right-limit, one can define the **derivative from the left** and the **derivative from the right**, respectively.

It is of great interest to consider different interpretations of the concept of derivative, and can prove to be useful in order to understand it better, from many angles. We here provide a list of some of such interpretations, without pretending to be mathematically rigorous. Look at them as (more or less) intuitive ways to think of the derivative of a function, and give yourself time to digest all of them.

The derivative can be thought of as:

1. Infinitesimal: the ratio of the infinitesimal change in the value of a function to the infinitesimal change in a function.
2. Logical: $f'(x) = d$ if and only if for every ϵ there is a δ such that when $0 < |\Delta| < \delta$,

$$\left| \frac{f(x + \Delta x) - f(x)}{\Delta x} - d \right| < \delta.$$

3. Geometric: the derivative is the slope of a line tangent to the graph of the function, if the graph has a tangent.

4. Rate: the instantaneous speed of $f(t)$, when t is time.
5. Approximation: The derivative of a function is the best linear approximation to the function near a point.
6. Microscopic: The derivative of a function is the limit of what you get by looking at it under a microscope of higher and higher power.

Definition 5.1.2: Higher Derivatives

Let $f : D \rightarrow \mathbb{R}$ be a function. We define the **higher derivatives** of f , if they exist, by

$$f^{(0)} = f, f^{(1)} = f', \dots, f^{(n+1)} = \left(f^{(n)}\right)',$$

for all $n \in \mathbb{N}$. If $f^{(n)}$ exists, for some $n \in \mathbb{N}$, f is called **n -times differentiable**. If the n -th derivative is also continuous, we say that f is **n -times continuously differentiable**.

We denote the set of n -times continuously differentiable functions on D by $C^n(D)$.

Definition 5.1.3: Smooth Functions

We define

$$C^\infty := \bigcap_{n=0}^{\infty} C^n(D),$$

and call functions $f \in C^\infty(D)$ **smooth** or **of class C^∞** .

5.1.2 Differentiation Rules

When we talked about continuity for the first time, we did by introducing the old good formal definition, the one with “epsilon” and “delta”. Nevertheless, when we need to handle functions and study their continuity properties, we usually don’t use the definition directly, but we rather use some rules (that sometimes look like tricks) to get around the epsilons and the deltas, so that our life can become easier. What we do here is provide some tricks and rules that we can use to study the differentiability.

Proposition 5.1.1: Derivative of Sum and Product

Let $D \subseteq \mathbb{R}$ and $x_0 \in D$ be an accumulation point of $D \setminus \{x_0\}$. Let $f, g : D \rightarrow \mathbb{R}$ be differentiable at x_0 . Then $f + g$ and $f \cdot g$ are also differentiable at x_0 , and the followings hold:

$$(f + g)'(x_0) = f'(x_0) + g'(x_0),$$

$$(fg)'(x_0) = f'(x_0)g(x_0) + f(x_0)g'(x_0).$$

This result can be easily extended to the case of higher-order derivatives. Assuming that f, g are both n -times differentiable at x_0 for some $n \in \mathbb{N}$, we have

$$(f + g)^{(n)}(x_0) = f^{(n)}(x_0) + g^{(n)}(x_0),$$

$$(fg)^{(n)}(x_0) = \sum_{k=0}^n \binom{n}{k} f^{(k)}(x_0)g^{(n-k)}(x_0).$$

Corollary 5.1.1: Derivatives of Polynomials

Polynomial functions are differentiable on all $\mathbb{R}$. And we have

$$(1)' = 0, \text{ and } (x^n)' = nx^{n-1}, \text{ for all } n \geq 1.$$

Theorem 5.1.1: Chain Rule

Let $D, E \subseteq \mathbb{R}$ be subsets, and let $x_0 \in D$ be an accumulation point of $D \setminus \{x_0\}$. Let $f : D \rightarrow \mathbb{R}$ be differentiable at x_0 such that $y_0 := f(x_0)$ is an accumulation point of $E \setminus \{y_0\}$. If $g : E \rightarrow \mathbb{R}$ is differentiable at y_0 , then $g \circ f : D \rightarrow \mathbb{R}$ is differentiable at x_0 , and

$$(g \circ f)'(x_0) = g'(f(x_0)) f'(x_0).$$

Corollary 5.1.2: Quotient Rule

Let $D \subseteq \mathbb{R}$, $x_0 \in D$ be an accumulation point of $D \setminus \{x_0\}$, and $f, g : D \rightarrow \mathbb{R}$ be differentiable at x_0 . If $g(x_0) \neq 0$, then $\frac{f}{g}$ is also differentiable at x_0 , and

$$\left(\frac{f}{g}\right)'(x_0) = \frac{f'(x_0)g(x_0) - f(x_0)g'(x_0)}{g(x_0)^2}.$$

Theorem 5.1.2: Derivative of the Inverse

Let $D, E \subseteq \mathbb{R}$, and let $f : D \rightarrow E$ be a continuous bijective mapping whose inverse $f^{-1} : E \rightarrow D$ is also continuous. Let $\bar{x} \in D$ be an accumulation point of $D \setminus \{\bar{x}\}$, and assume that f is differentiable at $\bar{x} \in D$, with $f'(\bar{x}) \neq 0$. Then f^{-1} is differentiable at $\bar{y} := f(\bar{x})$, and

$$(f^{-1})'(\bar{y}) = \frac{1}{f'(\bar{x})}.$$

As a remark, we want to highlight the fact that, in the previous expression for the derivative of f^{-1} , the derivative on the right-hand side and the one on the left-hand side are not evaluated at the same point, but at $\bar{y}$ and its preimage, respectively.

5.2 Main Theorems of Differential Calculus

The concept of a function's derivative is not only interesting *per se*, but it proves to be extremely useful in the understanding of the behavior of a function. For example, it shouldn't surprise that, given a function, it is often of interest to know whether it admits extreme points or not. Looking at the several interpretations of the derivatives given above, one can then easily hint that looking at the derivative of the function under consideration (if it exists) can help to find such extreme points.

To begin with, we need to formalize the idea of extreme points.

Definition 5.2.1: Local Extrema

Let $D \subseteq \mathbb{R}$, and $x_0 \in D$. We say that a function $f : D \rightarrow \mathbb{R}$ has a **local maximum** at x_0 if there exists $\delta > 0$ such that $f(x) \leq f(x_0)$ for all $x \in (x_0 - \delta, x_0 + \delta)$. If this inequality is strict, we say that f has an **isolated local maximum** at x_0 . The value $f(x_0)$ is called a **local maximal value** of f . A **local minimum**, an **isolated local minimum**, and a **local minimal value** of f are defined analogously.

Furthermore, we call x_0 a **local extremum** of f , and $f(x_0)$ a **local extremal value** of f , if f has a local minimum or a local maximum at x_0 .

Proposition 5.2.1: Local Extrema vs First Derivative

Let $D \subseteq \mathbb{R}$, and $f : D \rightarrow \mathbb{R}$. Suppose f is differentiable at a local extremum $x_0 \in D$, and x_0 is both a right-hand side and a left-hand side accumulation point of D . Then, we have $f'(x_0) = 0$.

Corollary 5.2.1: Local Extrema in an Interval

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$. Let $x_0 \in I$ be a local extremum of f . Then at least one of the following statements is true.

1. $x_0 \in I$ is an endpoint of I .
2. f is not differentiable at x_0 .
3. f is differentiable at x_0 and $f'(x_0) = 0$.

In particular, all local extrema of a differentiable function on an open interval are zeros of the derivative.

5.2.1 The Mean Value Theorem

Theorem 5.2.1: Rolle's Theorem

Let $a < b$, and $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function, differentiable on the open interval (a, b) . If $f(a) = f(b)$ holds, then there exists $\xi \in (a, b)$ with $f'(\xi) = 0$.

Theorem 5.2.2: Mean Value Theorem

Let $a < b$, and let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function, differentiable on the open interval (a, b) . Then there exists $\xi \in (a, b)$ with

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Theorem 5.2.3: Cauchy Mean Value Theorem

Let $a < b$, and let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions, both of them differentiable on the open interval (a, b) . Then there exists $\xi \in (a, b)$ with

$$g'(\xi) (f(b) - f(a)) = f'(\xi) (g(b) - g(a)).$$

If in addition $g'(x) \neq 0$ holds for all $x \in (a, b)$, then $g(a) \neq g(b)$, and

$$\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

5.2.2 L'Hôpital's Rule

The family of results known collectively as *Rule of l'Hôpital* (or l'Hospital) is named after the French mathematician and nobleman Guillaume François Antoine, Marquis de l'Hôpital. In fact, yet, it probably goes back to his personal teacher Johann Bernoulli, who agreed to let his student publish the results upon being fairly paid for his teachings.

Theorem 5.2.4: L'Hôpital's Rule

Let $a < b$, and $f, g : (a, b) \rightarrow \mathbb{R}$ differentiable functions. Suppose that:

1. $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$;
2. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$;
3. the limit $L := \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)}$ exists.

Then the limit $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)}$ also exists and is equal to L .

Theorem 5.2.5: L'Hôpital's Rule for Improper Limits

Let $a < b$, and $f, g : (a, b) \rightarrow \mathbb{R}$ differentiable functions. Suppose that:

1. $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (a, b)$;
2. $\lim_{x \rightarrow a^+} |f(x)| = \lim_{x \rightarrow a^+} |g(x)| = \infty$;
3. the limit $L := \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)}$ exists.

Then the limit $\lim_{x \rightarrow a^+} \frac{f(x)}{g(x)}$ also exists and is equal to L .

Theorem 5.2.6: L'Hôpital's Rule at Infinity

Let $R > 0$, and $f, g : (R, \infty) \rightarrow \mathbb{R}$ be differentiable functions. Suppose that

1. $g(x) \neq 0$ and $g'(x) \neq 0$ for all $x \in (R, \infty)$;
2. either $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} g(x) = 0$ or $\lim_{x \rightarrow \infty} |f(x)| = \lim_{x \rightarrow \infty} |g(x)| = \infty$;
3. the limit $L := \lim_{x \rightarrow a^+} \frac{f'(x)}{g'(x)}$ exists.

Then the limit $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$ also exists and is equal to L .

5.2.3 Monotonicity and Convexity via Differential Calculus

To simplify the notation, given a differentiable function $f : I \rightarrow \mathbb{R}$ defined on an interval $I \subseteq \mathbb{R}$, we will write $f' \geq 0$ meaning that $f'(x) \geq 0$ for all $x \in I$.

Moreover, when we speak of an interval I , we always assume it is non-empty.

Proposition 5.2.2: Monotonicity vs First Derivative

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be a differentiable function. Then

$$f' \geq 0 \Leftrightarrow f \text{ is increasing.}$$

Corollary 5.2.2: Constant Functions vs First Derivative

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$. Then f is constant if and only if it is differentiable and $f'(x) = 0$ holds for all $x \in I$.

Now, we introduce a new class of functions, namely the convex functions. Again, we can rely on some visual interpretation by looking at the graph of a (possibly nicely behaving) convex function. Roughly speaking, a function is said to be convex if, chosen two points $x_1 < x_2$ of its domain, its graph between the two points lays all under the segment that joints x_1 and x_2 .

Definition 5.2.2: Convex Functions

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$. Then f is called **convex** if, for all $a, b \in I$ with $a < b$ and all $t \in (0, 1)$, the inequality

$$f((1-t)a + tb) \leq (1-t)f(a) + tf(b)$$

holds. We say that f is **strictly convex** if the previous inequality is strict.

A function $g : I \rightarrow \mathbb{R}$ is called **(strictly) concave** if $-g$ is (strictly) convex.

Proposition 5.2.3: Convexity vs Monotonicity of First Derivative

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be a differentiable function. Then f is convex if and only if f' is increasing.

Corollary 5.2.3: Convexity vs Second Derivative

Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be a twice differentiable function. Then f is convex if and only if $f'' \geq 0$.

Chapter 6

The Riemann Integral

6.0.1 Decompositions and Step Functions

Definition 6.0.1: Decomposition of an Interval

A **decomposition** of $[a, b]$ is a finite set of points

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b,$$

with $n \in \mathbb{N}$. The points $x_0, \dots, x_n \in [a, b]$ are called the **division points** of the decomposition.

Given the decompositions $x_0, \dots, x_n$ and $y_0, \dots, y_m$, we call the latter a **refinement** of the former if the inclusion

$$\{x_0, \dots, x_n\} \subseteq \{y_0, \dots, y_m\}$$

holds. Note that the notion of refinement leads to an order relation on the set of all decompositions of $[a, b]$.

Definition 6.0.2: Step Functions

A function $f : [a, b] \rightarrow \mathbb{R}$ is called a **step function** if there exists a decomposition $a = x_0 < x_1 < \dots < x_n = b$ of $[a, b]$ such that for $k = 1, 2, \dots, n$ the restriction of f to the open interval (x_{k-1}, x_k) is constant.

The set of all the step functions (on a fixed closed interval, say $[a, b]$) constitutes a so-called linear space, as the next proposition shows.

Proposition 6.0.1: Linearity of the Space of Step Functions

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be step functions, and $\alpha, \beta \in \mathbb{R}$. Then $\alpha f + \beta g$ is also a step function.

6.0.2 The Integral of a Step Function

As you know, we are here trying to formalize the idea of computing the area of the region of space between the x -axis and the graph of a given function. The end-line of this formalization journey is the Riemann integral, and the very first step toward the goal is the integral of a step function. Since a step function is what we can also call a piece-wise constant function (over a finite interval), its integral will turn out to be the sum of a finite number of terms, each of which is equal to the area of a certain rectangle (taken with the suitable sign).

Definition 6.0.3: Integral of Step Functions

Let $f : [a, b] \rightarrow \mathbb{R}$ be a step function with respect to the decomposition $a = x_0 < x_1 < \dots < x_n = b$ of $[a, b]$. We define the **integral** of f on $[a, b]$ as the real number

$$\int_a^b f(x) dx := \sum_{k=1}^n c_k (x_k - x_{k-1}),$$

where c_k denotes the value of f on the interval (x_k, x_{k-1}) , $k = 1, 2, \dots, n$.

Proposition 6.0.2: Linearity of Integral of Step Functions

Let $g, f : [a, b] \rightarrow \mathbb{R}$ be step functions, and let $\alpha, \beta \in \mathbb{R}$. Then

$$\int_a^b (\alpha f + \beta g)(x) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

Proposition 6.0.3: Monotonicity for Integral of Step Functions

Let $g, f : [a, b] \rightarrow \mathbb{R}$ be step functions such that $f \leq g$. Then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

6.1 Definition and First Properties of the Riemann Integral

Before we start, let's simplify the notation by introducing the space

$$\mathcal{SF} := \{\text{step functions on } [a, b]\}.$$

6.1.1 Integrability of Real-valued Functions

Definition 6.1.1: Lower and Upper Sums

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function. Then we define the set of **lower sums** $\mathcal{L}(f) \subseteq \mathbb{R}$ and **upper sums** $\mathcal{U}(f) \subseteq \mathbb{R}$ of f by

$$\mathcal{L}(f) := \left\{ \int_a^b l \, dx \text{ such that } l \in \mathcal{SF} \text{ and } l \leq f \right\},$$

$$\mathcal{U}(f) := \left\{ \int_a^b u \, dx \text{ such that } u \in \mathcal{SF} \text{ and } u \geq f \right\},$$

respectively.

Definition 6.1.2: Riemann Integral

A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is called **Riemann integrable** if $\sup \mathcal{L}(f) = \inf \mathcal{U}(f)$. In this case, this common value is called the **Riemann integral** of f , and is expressed as

$$\int_a^b f \, dx = \sup \mathcal{L}(f) = \inf \mathcal{U}(f).$$

Proposition 6.1.1: Riemann Integrability Condition

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded. The function f is Riemann integrable if and only if for every $\epsilon > 0$ there exist step functions l and u such that

$$l \leq f \leq u \quad \text{and} \quad \int_a^b (u - l) \, dx < \epsilon.$$

In such a case, we have

$$\left| \int_a^b f \, dx - \int_a^b l \, dx \right| < \epsilon,$$

and

$$\left| \int_a^b f \, dx - \int_a^b u \, dx \right| < \epsilon.$$

6.1.2 Riemann Integrability Condition

We hope you remember that above we provided two results on the linearity and monotonicity of the integral. In that case, we were dealing with step functions, which are, you will agree, quite nice in their form, and thus it is not surprising that for them some nice properties hold. What we're going to do next is to show that, in fact, those nice properties are transferred to any Riemann integrable function. The intuitive reason why this holds true comes directly from the construction of the Riemann integral. As you have seen, the integral of a Riemann integrable function f is given as the supremum of a set of integrals of step functions that approximate f from below, or, equivalently, as the infimum of a set of integrals of step functions that approximate f from above. Hence, what the next theorems say is that the properties of linearity and monotonicity “survive” this approximation process.

Theorem 6.1.1: Linearity of the Riemann Integral

If $g, f : [a, b] \rightarrow \mathbb{R}$ are integrable, and $\alpha, \beta \in \mathbb{R}$. Then $\alpha f + \beta g$ is integrable, and

$$\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx.$$

Proposition 6.1.2: Monotonicity of the Riemann Integral

Let $g, f : [a, b] \rightarrow \mathbb{R}$ be integrable, and assume that $f \leq g$. Then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx.$$

Theorem 6.1.2: Triangle Inequality for Riemann Integral

Let $f : [a, b] \rightarrow \mathbb{R}$ be an integrable function. Then f^+, f^- , and $|f|$ are also integrable, and

$$\left| \int_a^b f dx \right| \leq \int_a^b |f| dx.$$

6.2 Integrability Theorems

6.2.1 Integrability of Monotone Functions

In the following discussion, we consider monotone functions on $[a, b]$. Before we start, note that a monotone function on a compact interval is necessarily bounded.

Theorem 6.2.1: Monotone Functions are Integrable

Every monotone function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

Definition 6.2.1: Piecewise Monotone Functions

A function $f : [a, b] \rightarrow \mathbb{R}$ is called **piecewise monotone** if there is a decomposition $a = x_0 < x_1 < \dots < x_n = b$ of $[a, b]$ such that $f|_{(x_{k-1}, x_k)}$ is monotone for all $k \in \{1, \dots, n\}$.

As the monotone functions, even the piecewise monotone functions (on a compact interval) are bounded, and we have the following generalization of the previous theorem.

Corollary 6.2.1: Piecewise Monotone Functions are Integrable

Every piecewise monotone function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

6.2.2 Integrability of Continuous Functions

Theorem 6.2.2: Continuous functions are integrable

Every continuous function $f : [a, b] \rightarrow \mathbb{R}$ is integrable.

As for the monotone functions, we can generalize the previous statement to the class of piecewise continuous functions.

Definition 6.2.2: Piecewise Continuous Functions

A function $f : [a, b] \rightarrow \mathbb{R}$ is called **piecewise continuous** if there is a decomposition $a = x_0 < x_1 < \dots < x_n = b$ of $[a, b]$ such that $f|_{(x_{k-1}, x_k)}$ is continuous for all $k \in \{1, \dots, n\}$, and both the limits $\lim_{x \rightarrow x_{k-1}^+} f(x)$ and $\lim_{x \rightarrow x_k^-} f(x)$ exist.

Corollary 6.2.2: Piecewise Continuous Functions are Integrable

Every piecewise continuous function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable.

6.2.3 Integration and Sequences of Functions

Earlier in these notes, we introduced the concepts of pointwise and uniform convergence for a sequence $(f_n)_{n \in \mathbb{N}}$ of functions. We then said that the latter is stronger than the former, and if the functions f_n are continuous and converge to some function f , then also f is continuous. A similar result holds in the context of integration, as we shall see soon.

Theorem 6.2.3: Uniform Convergence and Riemann Integral Commute

Let $(f_n)_{n \in \mathbb{N}}$ with $f_n : [a, b] \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, be a sequence of integrable functions converging uniformly to a function $f : [a, b] \rightarrow \mathbb{R}$. Then f is also integrable, and

$$\int_a^b f \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n \, dx.$$

We think it is important to look carefully at this result. It may seem to be trivial and straightforward, but this is far from the truth. Later in your studies, you will realize how careful one should be in swapping the integral sign with the one of limit. This is in general not possible, and thus uniform convergence proves to be a remarkable property for a sequence of functions.

Chapter 7

The Derivative and the Riemann Integral

A priori, the last two chapters may look unrelated to each other. The following discussion aims to unravel a deep connection between the two main concepts discussed there, namely those of derivative and integral.

7.1 The Fundamental Theorem of Calculus

7.1.1 The Fundamental Theorem

Definition 7.1.1: Primitive Function

Let $I \subset \mathbb{R}$ be an interval, and let $f : I \rightarrow \mathbb{R}$ be a function. Any differentiable function $F : I \rightarrow \mathbb{R}$ such that $F' = f$ is called a **primitive** of f .

We say “a” primitive function because, in general, it is not unique. More precisely, a primitive function needs not exist. Could you provide any examples? A common one is given by the function $\text{sign}(x)$, defined by

$$\text{sign}(x) := \begin{cases} -1, & \text{if } x < 0, \\ 0, & \text{if } x = 0, \\ +1, & \text{if } x > 0. \end{cases}$$

As for the non-uniqueness, consider the function $f \equiv 0$. Clearly, any constant function F is a primitive of f .

Theorem 7.1.1: Fundamental Theorem of Calculus

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then, for every $C \in \mathbb{R}$, the function $F : [a, b] \rightarrow \mathbb{R}$ defined by

$$F(x) := \int_a^x f(t) dt + C$$

is a primitive of f .

Moreover, Any primitive of f has exactly this form for some constant $C \in \mathbb{R}$.

Note that the condition required on f is the one of continuity, which suffices to make F differentiable. In particular, this means that the primitive of a continuous function (on a closed interval $[a, b]$) is more regular than the original function, which needs not be differentiable. We will elaborate more on this question soon.

Corollary 7.1.1: Integral vs Derivative

Let $F : [a, b] \rightarrow \mathbb{R}$ be a continuously differentiable function. Then, for all $x \in [a, b]$, we have

$$F(x) = F(a) + \int_a^x F'(t) dt.$$

This is a direct consequence of the previous theorem. It suffices to observe that F is trivially a primitive of F' .

Corollary 7.1.2: Riemann Integral and Primitives

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous, and let $F : [a, b] \rightarrow \mathbb{R}$ be a primitive of f . Then, we have

$$\int_a^b f(t) dt = F(b) - F(a).$$

To see the power of these results in action, you could do the following useful exercise (provided in the lecture notes). Given a function $f : [a, b] \rightarrow \mathbb{R}$, assume it is discontinuous at most finitely many points. Show that $F(x) := \int_a^x f(t) dt$ is continuous in $[a, b]$, differentiable wherever f is continuous, and at such points it satisfies $F'(x) = f(x)$.

This exercise leads to an interesting remark. In general, if we take the derivative of a (differentiable) function f , then the function f' is “less regular” than f . For example, you can consider

$$f(x) := \begin{cases} x^2 \cdot \sin(1/x) & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

In this case, f is differentiable (thus continuous), but its first derivative is discontinuous (at 0), and thus not even differentiable.

When we integrate a function, yet, we are presented with the opposite situation. As shown by the previous exercise, the primitive function F can be continuous even if f has (some) discontinuities. More generally, by integrating, we gain some regularity property.

7.1.2 Integration by Parts and by Substitution

Before we start, let's recall some notation:

$$[h(x)]_a^b := h(b) - h(a).$$

Theorem 7.1.2: Integration by Parts

Let $g, f : [a, b] \rightarrow \mathbb{R}$ be differentiable functions. Then, we have

$$\int_a^b f(x)g'(x) dx = [f(x)g(x)]_a^b - \int_a^b f'(x)g(x) dx. \quad (7.1)$$

To remember this result more easily, think of the formula for the derivative of the product of two functions. As a convention, we set

$$\int_b^a h(x) dx := - \int_a^b h(x) dx.$$

By doing so, we are no longer constrained to check whether the extremes of integration are in increasing order. Moreover, if we look at this equation, we see that to pass from one side to the other, it suffices to add a minus sign. This is not simply a reasonable convention, but one can justify the equality by calling upon the theory of orientation of manifolds, in the broader context of differential geometry.

Theorem 7.1.3: Integration by Substitution, 1st Form

Let $I, J \subseteq \mathbb{R}$ be intervals, and let $f : I \rightarrow \mathbb{R}$, $g : J \rightarrow \mathbb{R}$ be differentiable and continuous, respectively. Then, for any $[a, b] \subseteq I$, we have

$$\int_a^b g(f(x))f'(x) dx = \int_{f(a)}^{f(b)} g(y) dy. \quad (7.2)$$

Theorem 7.1.4: Integration by Substitution, 2nd Form

Let $I, J \subseteq \mathbb{R}$ be intervals, and let $f : I \rightarrow \mathbb{R}$, $g : J \rightarrow \mathbb{R}$ be continuously differentiable and continuous, respectively. Given $[a, b] \subseteq I$, assume that $f'(x) \neq 0$ for all $x \in [a, b]$, and let f^{-1} be the inverse of $f|_{[a, b]}$. Then, we have

$$\int_a^b g(f(x)) dx = \int_{f(a)}^{f(b)} g(y) (f^{-1})'(y) dy. \quad (7.3)$$

7.1.3 Improper Integrals

So far, we have made sense of integrating a function over a compact interval. In the following, we'd like to extend the concept of integrals, allowing the set over which we integrate to be unbounded. In parallel, we will be able to define the integral of an unbounded function on a suitable compact interval.

Given a non-empty interval $I \subseteq \mathbb{R}$ (possibly unbounded), we say that a function $f : I \rightarrow \mathbb{R}$ is **locally integrable** if $f|_{[a, b]}$ is integrable for any choice of $a, b \in \mathbb{R}$ with $a < b$, and $[a, b] \subseteq I$.

Definition 7.1.2: Improper Integrals

Let $I \subseteq \mathbb{R}$ be an interval, and $f : I \rightarrow \mathbb{R}$ be a locally integrable function. Set $c := \inf(I) \in \mathbb{R} \cup \{-\infty\}$, and $d := \sup(I) \in \mathbb{R} \cup \{\infty\}$, and choose arbitrarily $x_0 \in I$. We define the **improper integral** of f on I as

$$\int_c^d f(x) dx = \lim_{a \rightarrow c^+} \int_a^{x_0} f(x) dx + \lim_{b \rightarrow d^-} \int_{x_0}^b f(x) dx$$

whenever both limits exist and the sum makes sense (so, if the limits are infinite, we do not admit an expression of the form $\infty - \infty$).

If the limit is finite, we say that the improper integral converges. If the limit is equal to ∞ or $-\infty$, we say that the improper limit diverges. Otherwise, we call the improper integral not convergent.

Lemma 7.1.1: Improper Integral of Non-negative Functions

Let $a \in \mathbb{R}$, and $f : [a, \infty) \rightarrow \mathbb{R}_0^+$ be a non-negative locally integrable function. Then, we have

$$\int_a^\infty f(x) dx = \sup \left\{ \int_a^b f(x) dx \text{ such that } b > a \right\}.$$

Note that we could have instead assumed that f was non-positive, and adjusted the signs in the expression given above.

You may have already noticed that the concept of integral can be thought of as a generalization of the one of series in a continuous setup. The following result makes the relationship between the two concepts quantitative.

Theorem 7.1.5: Integral Test for Series

Let $f : [0, \infty) \rightarrow \mathbb{R}_0^+$ be a monotonically decreasing function. Then, for each $N \in \mathbb{N}$, we have

$$\sum_{n=1}^{N+1} f(n) \leq \int_0^{N+1} f(x) dx \leq \sum_{n=0}^N f(n).$$

In particular, we have

$$\sum_{n=1}^{\infty} f(n) \leq \int_0^{\infty} f(x) dx \leq \sum_{n=0}^{\infty} f(n).$$

Therefore, being the term $f(0)$ negligible in the study of the convergence of a series, we see that the series $\sum_{n=1}^{\infty} f(n)$ converges if and only if the improper integral $\int_0^{\infty} f(x) dx$ is convergent.

7.2 Integration and Differentiation of Power Series

Theorem 7.2.1: Integration of Power Series

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with radius of convergence $R > 0$. Then the power series

$$F(x) := \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$$

has also radius of convergence R , and is a primitive of f on $(-R, R)$.

Do you remember that $(x^n)' = nx^{n-1}$? Well, if you look at the expression of F in the previous statement, you should see that this function is obtained by integrating the series that defines f term-by-term. So, we see that a power series can be integrated term-by-term, and by doing so one gets a primitive of the original series on the ball centered in 0 and with radius the one of convergence.

Now that we know the intimate relationship between integration and differentiation, we should not be surprised by the following result, which is, in some sense, the counterpart of the previous.

Corollary 7.2.1: Differentiation of Power Series

Let $f(x) = \sum_{n=0}^{\infty} a_n x^n$ be a power series with a radius of convergence $R > 0$. Then $f : (-R, R) \rightarrow \mathbb{R}$ is differentiable, and we have

$$f'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}, \text{ for all } x \in (-R, R),$$

where the power series on the right-hand side has also radius of convergence R .

As you may have noticed, the power series that defines f' starts with the term corresponding to $n = 1$. This is easily justified, upon noticing that this power series can be obtained from the original one by differentiation term-by-term. Thus, being the term $a_0 x^0$ a constant, its first derivative is null.

7.3 Integration Methods

This section is supposed neither to be complete nor exhaustive. Rather, it must be intended as a collection of the main methods of integration discussed during the course, at which one can look when in search of help to compute some integral, and to refresh their mind in case they are not able to recall what are the options at their disposal. Before we start, let $I \subseteq \mathbb{R}$ be an interval, and $f : I \rightarrow \mathbb{R}$ a function. By the notation

$$\int f(x) dx = F(x) + C,$$

we mean that F is a primitive function of f . The constant C is read as an indefinite constant, called **integration constant**. Since I is an interval (thus, a connected set), any two primitive functions of f on I differ by a constant. We call $F(x) + C$ the **indefinite integral** of f . The methods we list in the following are, indeed, aimed at determining indefinite integrals.

1. Integration by parts.

The essence of this method is all given by the expression 7.1.

2. Integration by substitution.

As for this second method, it relies on expressions 7.2 and 7.3.

3. Integration of rational functions.

7.4 Taylor Series

Let $D \subseteq \mathbb{R}$ be an open interval, and let $f : D \rightarrow \mathbb{C}$ be an n times differentiable function. The n -th **Taylor approximation** of f around a point $x_0 \in D$ is the polynomial function

$$P_n(x) := \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k.$$

By a direct computation, one can see that

$$P_n^{(k)}(x_0) = f^{(k)}(x_0),$$

for all $k \in \{0, \dots, n\}$.

As the name suggests, P_n is an approximation of f . The following results aim to quantify the goodness of this approximation. In particular, they show that the closer one gets to the point x_0 , the more precise the approximation is. Moreover, we need to note that an error always exists in approximating a function with its Taylor approximation, unless the given function is a polynomial (or the value to compute is the one at x_0).

One further remark is about how the Taylor approximation is obtained. We have seen in the previous chapters that the study of the behavior of a function can be undertaken by looking at its derivatives (when they exist), and our comprehension of the behavior enriches whenever we go from some derivative to the successive. Well, what we do with the n -th Taylor approximation is to take all the information given by the first n derivatives of a f at some point x_0 , and encapsulate this into a polynomial, namely P_n . Clearly, the higher n is, the more reliable the approximation will prove to be.

Theorem 7.4.1: Taylor Expansion to Order n with Integral Remainder

Let $n \geq 1$, $f : [a, b] \rightarrow \mathbb{R}$ a n -times continuously differentiable function, and fix $x_0 \in [a, b]$. Then, for all $x \in [a, b]$, we have

$$f(x) = P_{n-1}(x) + \int_{x_0}^x f^{(n)}(t) \frac{(x-t)^{n-1}}{(n-1)!} dt,$$

where P_{n-1} is the $(n-1)$ -th Taylor approximation of f .

Corollary 7.4.1: Taylor Approximation with Big-O

Let $n \geq 1$, $f : [a, b] \rightarrow \mathbb{R}$ a n -times continuously differentiable function, and fix $x_0 \in [a, b]$. Then, for all $x \in [a, b]$, we have

$$f(x) = P_{n-1}(x) + O(|x - x_0|^n), \text{ as } x \rightarrow x_0.$$

Corollary 7.4.2: Taylor Approximation with Little-o

Let $n \geq 1$, $f : [a, b] \rightarrow \mathbb{R}$ a n -times continuously differentiable function, and fix $x_0 \in [a, b]$. Then, for all $x \in [a, b]$, we have

$$f(x) = P_n(x) + o(|x - x_0|^n), \text{ as } x \rightarrow x_0.$$

Theorem 7.4.2: Taylor Expansion to Order n with Lagrange Remainder

Let $n \geq 1$, $f : [a, b] \rightarrow \mathbb{R}$ a n -times continuously differentiable function, and fix $x_0 \in [a, b]$. Then, for all $x \in [a, b]$, there exists $\xi_L \in (x_0, x)$ such that the **Lagrange remainder** formula holds:

$$f(x) - P_{n-1}(x) = \frac{1}{n!} f^{(n)}(\xi_L) (x - x_0)^n.$$

Theorem 7.4.3: Taylor Approximation with Big-O, 2nd version

Let $n \geq 1$, $f : [a, b] \rightarrow \mathbb{R}$ a n -times continuously differentiable function, and fix $x_0 \in [a, b]$. Assume that there exists $M > 0$ such that $|f^{(n)}(x)| \leq M$ for all $x \in [a, b]$. Then, we have

$$f(x) = P_{n-1}(x) + O(|x - x_0|^n), \text{ as } x \rightarrow x_0.$$

7.4.1 Analytic Functions

We said that “the higher n is, the more reliable the approximation will be”. So, it is natural to ask what happens to the error if one replaces the polynomial P_n with the **Taylor series**

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k,$$

and if there are any chances for the approximation error for the Taylor series to be equal to zero. Well, in general, neither the Taylor series can recover $f(x)$, and the functions that satisfy this property (i.e., they coincide with the infinite Taylor expansion) are rather special. So special that they deserve a definition.

Definition 7.4.1: Analytic Functions

Let $I \subseteq \mathbb{R}$ be an interval, and $x_0 \in I$. A smooth function $f : I \rightarrow \mathbb{R}$ is called **analytic** at x_0 if there exists $\delta > 0$ such that the Taylor series of f around x_0 has radius of convergence $R > \delta$ and

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n,$$

for all $x \in (x_0 - \delta, x_0 + \delta) \cap I$. We say that f is analytic in I if it is analytic at all points in I .

As said above, in general, a function cannot be recovered by its Taylor series. The next theorem provides a criterion that guarantees that the Taylor series of f converges to f in a neighborhood of some point x_0 .

Theorem 7.4.4: A Criterion for Analyticity at x_0

Let $I \subseteq \mathbb{R}$ be an interval, and $f : I \rightarrow \mathbb{R}$ a smooth function. Given $x_0 \in I$, assume that there exist constants $r, c, A > 0$ such that

$$|f^{(n)}(x)| \leq cA^n n! \quad \text{for all } x \in (x_0 - r, x_0 + r) \cap I, n \in \mathbb{N}.$$

Then f is analytic.

Corollary 7.4.3: A Criterion for Analyticity

Let $f : [a, b] \rightarrow \mathbb{R}$ be a smooth function, and assume there exist constants $c, A > 0$ such that

$$|f^{(n)}(x)| \leq cA^n n! \quad \text{for all } x \in [a, b], n \in \mathbb{N}.$$

Then f is analytic.